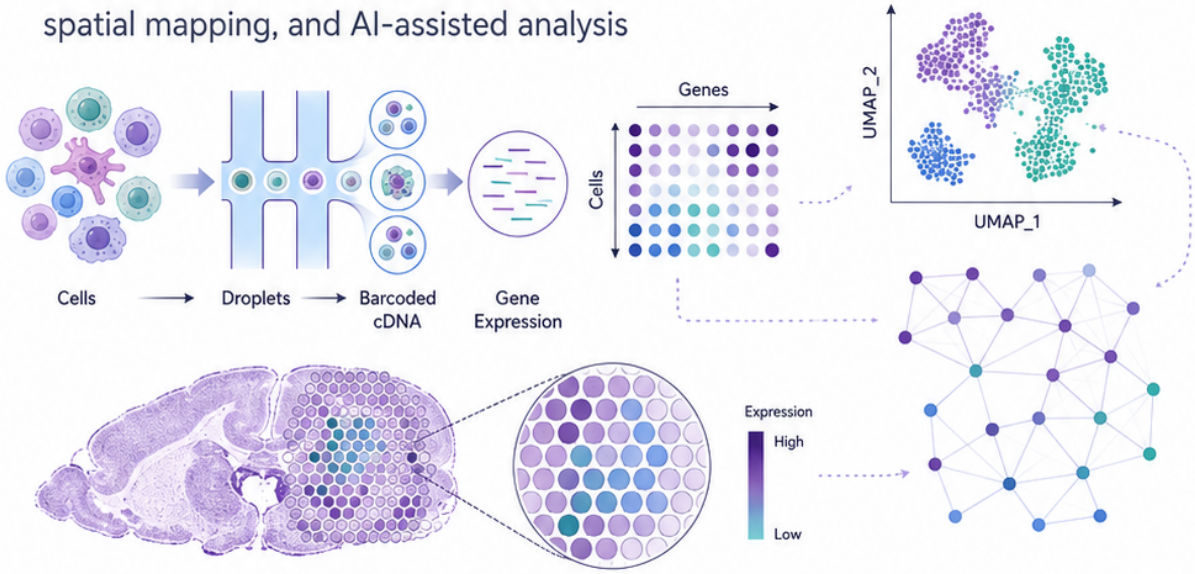
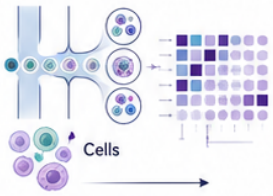
BIOINFORMATICS: FROM MULTI-OMICS DATA TO DISCOVERY

WEEK 8 READING MATERIAL

Single-Cell and Spatial Transcriptomics

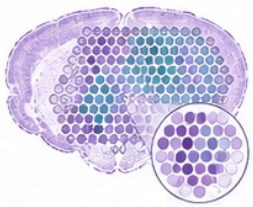
From cell states and tissue architecture to annotation, spatial mapping, and AI-assisted analysis





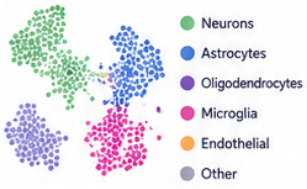
SINGLE-CELL RNA-SEQ

Profile thousands of cells to reveal heterogeneity and dynamic cell states.



SPATIAL TRANSCRIPTOMICS

Map gene expression in tissue context to uncover spatial organization and niches.



CELL-TYPE ANNOTATION

Annotate cell types using markers, references, and multimodal information.

AI-ASSISTED ANALYSIS

Leverage AI to denoise, annotate, visualize, and discover biological insights at scale.

Fall 2026 | Two 40-Minute Undergraduate Sessions | Dr. Liwei Xie

> ***About the Guest Lecturer — Prof. Guangchuang Yu (余光创):*** *Prof. Yu is a Professor and Principal Investigator at the Department of Bioinformatics, School of Basic Medical Sciences, Southern Medical University, Guangzhou. He is the author of some of the most widely used bioinformatics R packages in the world, including* ***clusterProfiler*** *(>15,000 citations),* ***ggplot2*** *extensions (ggtree, ggfun, aplot),* ***enrichplot****,* ***DOSE****, and* ***ChIPseeker****. His research focuses on bioinformatics software development, functional enrichment analysis, and phylogenetic visualization. The clusterProfiler package, which students will use in Week 9 for multi-omics pathway enrichment, was developed by Prof. Yu's laboratory. This week's handout has been designed in consultation with Prof. Yu's teaching philosophy: rigorous conceptual foundations combined with immediately applicable computational skills.*

> *How to Use This Handout: Week 8 combines core concepts with an AI-aware computational perspective. The two 40-minute sessions introduce single-cell and spatial transcriptomics from experimental design to replicate-aware statistics, foundation models, and inspectable AI-agent workflows. Students should treat AI as an accelerator for planning, coding, annotation, literature retrieval, and visualization - not as an authority for cell identity, spatial mechanism, or statistical validity. Read this handout before class and use the agent verification checklists during practice.*

> ***Conceptual Bridge from Week 7:*** *Proteomics and metabolomics (Week 7) measured the functional output of cells — what proteins are present and what metabolites are being produced. Both were* ***bulk measurements****: they averaged the signal across all cells in a sample. This averaging conceals a fundamental biological truth: not all cells in a tissue are the same. A liver biopsy contains hepatocytes, Kupffer cells, stellate cells, endothelial cells, and cholangiocytes — each with a distinct transcriptome, proteome, and metabolome.* ***Single-cell RNA sequencing (scRNA-Seq)*** *resolves this cellular heterogeneity at the level of individual cells.* ***Spatial transcriptomics*** *goes one step further, preserving the physical location of each cell within the tissue. Together, these technologies answer the question:* ***"Which cell type is doing what, and where in the tissue is it happening?"***

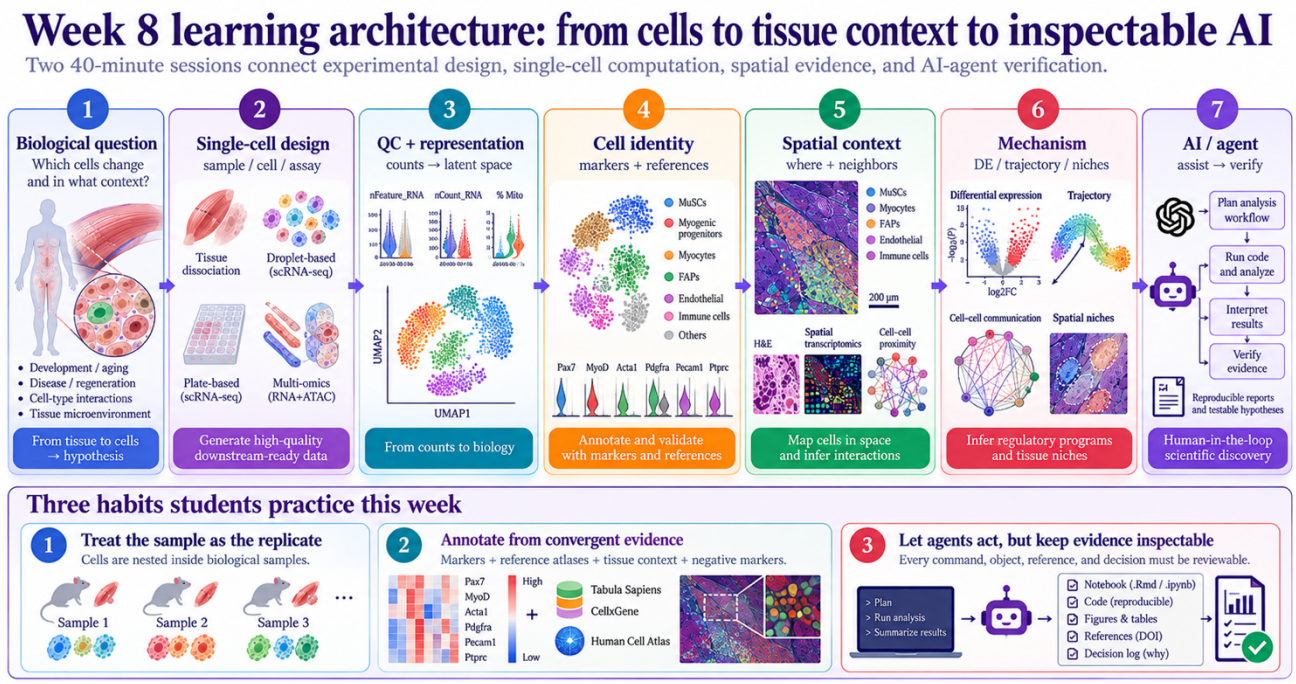


Figure 1. Week 8 learning architecture. The course moves from biological question and sample design to single-cell and spatial evidence, with AI/agent assistance embedded inside an explicitly verified workflow.

SESSION 1 (40 min): Single-Cell RNA Sequencing — Principles, Workflow, and Analysis

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1.1 Why Single-Cell Resolution? The Limitations of Bulk RNA-Seq

Every bulk omics measurement — whether RNA-Seq, proteomics, or metabolomics — produces a single average profile representing the population-weighted mean of all cells in the sample. This averaging is not merely a technical inconvenience; it is a fundamental limitation that prevents the detection of biologically critical phenomena. Three specific problems illustrate why single-cell resolution is scientifically necessary.

Problem 1: Rare Cell Detection. Consider a tumor biopsy containing 95% cancer cells and 5% tumor-infiltrating T cells. In bulk RNA-Seq, the T cell transcriptional signal is diluted 19-fold by the dominant cancer cell signal. Genes expressed exclusively in T cells will appear at 5% of their true expression level — potentially below the detection threshold. The entire T cell compartment, which is clinically critical for immunotherapy response prediction, is effectively invisible. scRNA-Seq detects and fully characterizes this 5% minority population with the same sensitivity as the majority population [1].

Problem 2: Cell-Type-Specific Responses. A drug may activate the NF-κB pathway in macrophages while simultaneously suppressing it in epithelial cells. In bulk RNA-Seq, these opposing effects cancel out, and the pathway appears unchanged. This masking of cell-type-specific responses is a major reason why bulk transcriptomics studies often fail to identify the mechanistic basis of drug effects or disease processes. scRNA-Seq resolves each cell type's response independently, revealing the full complexity of the biological response.

Problem 3: Developmental Trajectories. Cell differentiation is a continuous process: a hematopoietic stem cell (HSC) differentiates through a series of intermediate progenitor states before becoming a mature monocyte or T cell. At any given moment, a tissue contains cells at every stage of this continuum. Bulk RNA-Seq captures only the population average — a blurred composite of all stages. scRNA-Seq captures each cell's individual transcriptional state, enabling computational reconstruction of the full differentiation trajectory (pseudotime analysis, Section 1.4).

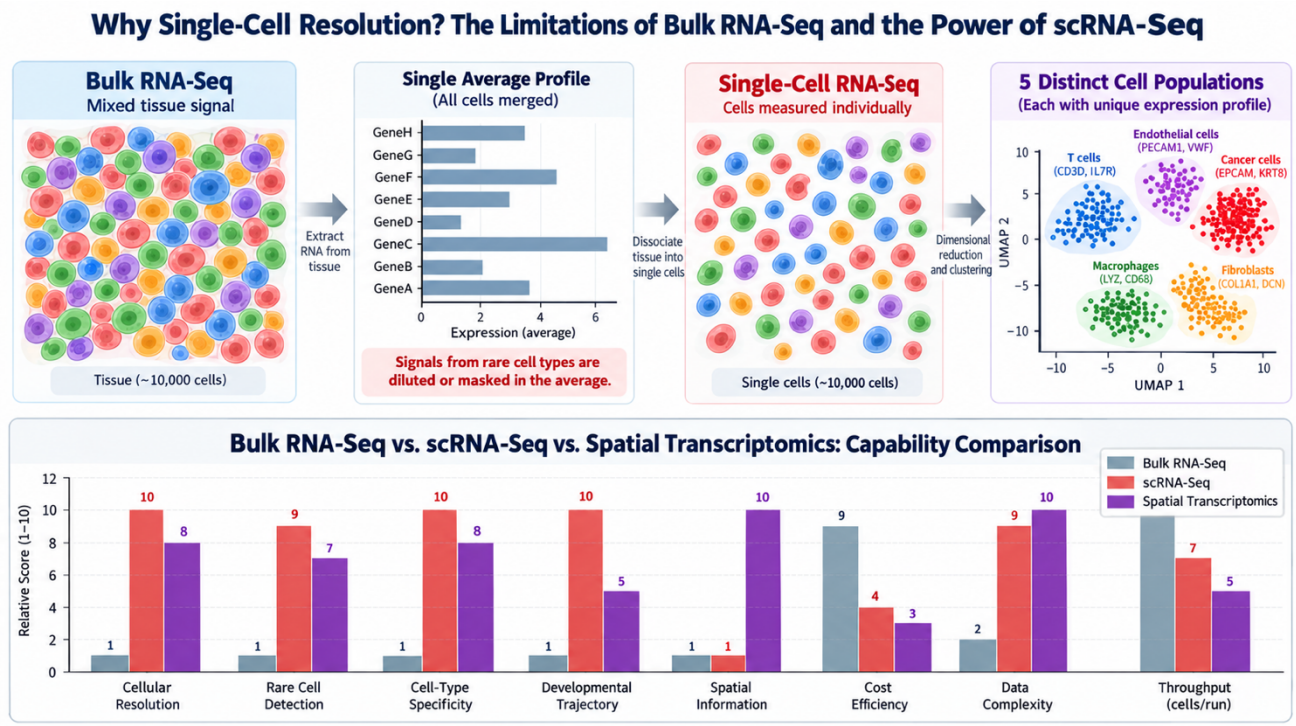

Figure 1. Top row: Bulk RNA-Seq (left) homogenizes all cells into a single average expression profile, masking the 2–5% rare cell populations. scRNA-Seq (right) of the same tissue resolves 5 distinct cell populations, each with a unique expression profile. Bottom panel: Capability comparison across 8 dimensions — scRNA-Seq leads in cellular resolution, rare cell detection, cell-type-specific DE, and developmental trajectory; spatial transcriptomics uniquely provides spatial information; bulk RNA-Seq retains advantages in cost efficiency and throughput.

Feature	Bulk RNA-Seq	scRNA-Seq	snRNA-Seq	Spatial Transcriptomics
Resolution	Population average	Single cell	Single nucleus	Spot (1–10 cells) or single cell
Input material	Total RNA from bulk tissue	Freshly dissociated single cells	Frozen tissue nuclei	Fresh-frozen or FFPE tissue section
Sensitivity (genes/cell)	~15,000–20,000 (bulk)	~2,000–5,000	~1,500–3,500	~2,000–5,000 (Visium)
Throughput (cells/run)	N/A (bulk)	500–50,000	500–50,000	~3,000–5,000 spots
Cost per cell	Very low (~$0.001)	Medium (~$0.10–0.50)	Medium (~$0.10–0.50)	High (~$1–5 per spot)
Spatial information	No	No	No	Yes
Fresh tissue required?	No	Yes (ideally)	No (frozen OK)	No (FFPE compatible)
Key software	DESeq2, edgeR, limma	Seurat, Scanpy	Seurat, Scanpy	Seurat, Squidpy, Giotto

Table 1. Bulk RNA-Seq vs. scRNA-Seq vs. snRNA-Seq vs. Spatial Transcriptomics: comprehensive comparison across 8 dimensions [1] [2].

1.2 scRNA-Seq Technology: From Cell to Count Matrix

1.2.1 The Three Cell Capture Strategies

Three fundamentally different approaches exist for capturing single cells for RNA sequencing, each with distinct tradeoffs in throughput, sensitivity, and cost.

Microfluidics-based droplet capture (10x Genomics Chromium) is the dominant platform, used in approximately 80% of published scRNA-Seq studies. The 10x Chromium Controller partitions cells into nanoliter-scale oil droplets called **GEMs (Gel Beads in Emulsion)**. Each GEM ideally contains one cell and one barcoded gel bead. The gel bead carries thousands of oligonucleotides, each containing a unique **cell barcode** (16 bp in 10x v3 chemistry) and a random **UMI** (Unique Molecular Identifier, 12 bp) attached to a poly-dT capture sequence. When the GEM is formed, the cell lyses, releasing its mRNA, which is captured by the poly-dT sequence and reverse-transcribed into cDNA — tagged with the cell barcode and UMI. After GEM breakage, all cDNA from all cells is pooled and amplified together, but each molecule retains its cell barcode, enabling computational assignment of reads to cells. A standard 10x v3.1 run captures approximately 500–10,000 cells per lane.

Combinatorial indexing (sci-RNA-seq3) avoids physical cell isolation entirely. Cells or nuclei are split into wells, each containing a unique DNA barcode that is ligated to all nucleic acids in that well. Cells are then pooled and split again into a second set of barcoded wells. The combinatorial combination of two (or three) barcodes uniquely identifies each cell. This approach can profile **millions of cells** in a single experiment at very low cost per cell, making it ideal for large-scale atlas projects. The tradeoff is lower sensitivity per cell (~1,000–2,000 genes detected) compared to droplet-based methods.

Plate-based methods (Smart-seq2/3) sort individual cells into wells of a 96- or 384-well plate using FACS. Each cell is processed individually, enabling full-length transcript coverage and higher sensitivity per cell (~5,000–8,000 genes). The major limitation is throughput: only 384 cells per plate. Smart-seq2/3 is best suited for rare cell populations, isoform analysis, or studies requiring the highest possible sensitivity.

1.2.2 Cell Barcodes and UMIs — The Two Key Innovations

A **cell barcode** is a short DNA sequence (16 bp in 10x v3, allowing 4^16 ≈ 4.3 billion unique sequences) that is unique to each gel bead and therefore to each cell captured in a GEM. Every cDNA molecule synthesized from a given cell carries the same cell barcode. After sequencing, computational demultiplexing assigns each read to its cell of origin by matching the cell barcode against a known whitelist of valid barcodes.

A UMI (Unique Molecular Identifier) is a short random sequence (12 bp in 10x v3) added to each mRNA molecule individually during reverse transcription, before PCR amplification. Because each original mRNA molecule receives a unique UMI, PCR duplicates of the same molecule all carry the same UMI and can be collapsed into a single count. Without UMIs, PCR amplification introduces 10–100 copies of each molecule, causing highly expressed genes to be dramatically over-counted.

With UMIs, each original mRNA molecule is counted exactly once, enabling deduplicated counting of captured molecules 1.

> *Worked Example: Suppose a cell expresses 100 molecules of Gene A. After reverse transcription, each receives a unique UMI. After 12 PCR cycles, each molecule produces ~4,000 copies, giving ~400,000 reads mapping to Gene A. Without UMI deduplication: count = 400,000 (4,000× overestimate). With UMI deduplication: all reads with the same cell barcode + gene + UMI are collapsed to one count, recovering 100 captured molecules in this idealized example; real experiments do not capture every transcript in the cell.*

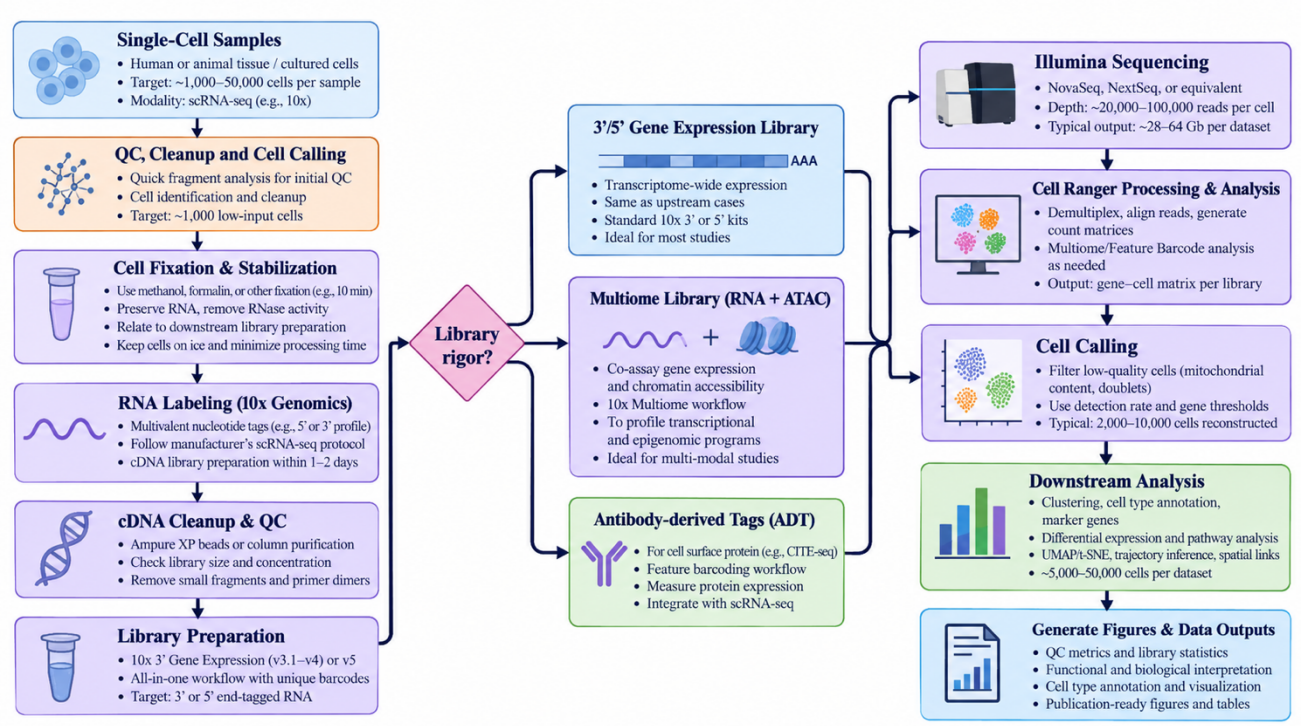

Figure 2. The complete 10x Genomics Chromium scRNA-Seq workflow, from single-cell suspension through GEM generation, reverse transcription with cell barcode and UMI tagging, library preparation, Illumina sequencing, Cell Ranger processing, and count matrix generation. The workflow branches at the library preparation step to accommodate Gene Expression, Multiome (ATAC+RNA), and CITE-seq (protein+RNA) applications.

Platform	Cells/Run	Cost/Cell	Genes Detected/Cell	Full-Length	Fresh/Fixed	Key Software
10x Chromium v3.1	500–10,000	~$0.10–0.30	~2,000–4,000	3' end only	Fresh	Cell Ranger, Seurat
10x Chromium v4 (FLEX)	500–10,000	~$0.15–0.35	~2,500–4,500	3' end only	Fixed (FFPE)	Cell Ranger, Seurat
Smart-seq3	96–384	~$5–15	~5,000–8,000	Full-length	Fresh	STARsolo, Seurat
sci-RNA-seq3	10,000–1,000,000	~$0.01–0.05	~1,000–2,000	3' end only	Fixed	STARsolo, Seurat
DNBelab C4 (BGI)	500–10,000	~$0.05–0.15	~2,000–3,500	3' end only	Fresh	CellRanger-DNBC4

Platform	Cells/Run	Cost/Cell	Genes Detected/Cell	Full-Length	Fresh/Fixed	Key Software
Parse Biosciences Evercode	10,000–1,000,000	~$0.02–0.10	~1,500–3,000	3' end only	Fixed	Parse pipeline, Seurat

Table 2. scRNA-Seq platform comparison: cells per run, cost per cell, sensitivity, coverage, sample type, and key software 1 2.

1.3 Bioinformatics Pipeline: Cell Ranger → Seurat → Biological Interpretation

1.3.1 Alignment and Count Matrix Generation

Raw FASTQ files are processed by **Cell Ranger** (10x Genomics) or **STARsolo** (open-source). Both perform: (1) cell barcode correction against the known whitelist; (2) read alignment to the reference genome (GRCh38); (3) UMI deduplication; and (4) gene counting per cell. The output is a **sparse count matrix** in MEX format — three files: matrix.mtx, barcodes.tsv, and features.tsv. A typical 10x run produces ~5,000 cells × ~33,000 genes, with approximately 95% zero entries — this extreme sparsity is a defining characteristic of scRNA-Seq data. **Cell calling** distinguishes real cells from empty droplets: Cell Ranger's **emptyDrops** algorithm uses a statistical test to identify barcodes with expression profiles significantly different from the ambient RNA background.

1.3.2 Quality Control — The Three Key Metrics

After loading the count matrix into Seurat, three QC metrics are universally applied 2:

nFeature_RNA (number of unique genes per cell): Too low (<200) = empty droplet or dead cell; too high (>6,000–8,000) = **doublet** (two cells in one GEM). **nCount_RNA** (total UMI count per cell): Correlated with nFeature_RNA; extreme outliers indicate doublets. **percent.mt** (percentage of reads mapping to mitochondrial genes): High percent.mt (>20–25%) indicates dying or stressed cells — when a cell's cytoplasmic membrane ruptures, cytoplasmic mRNA leaks out while mitochondria (membrane-enclosed) retain their mRNA, inflating the mitochondrial fraction. **DoubletFinder** and **Scrublet** provide dedicated doublet detection by simulating artificial doublets and identifying real cells with similar transcriptomes.

1.3.3 Normalization, Feature Selection, and Dimensionality Reduction

Library size normalization (NormalizeData() in Seurat) divides each cell's counts by total UMI × 10,000 (CP10K) and applies log1p transformation. SCTransform (regularized negative binomial regression) is the recommended alternative for datasets with high sequencing depth variation 3 — it models the mean-variance relationship per gene and uses Pearson residuals as normalized values. Important: do not sequentially normalize the same assay values with both methods for the same analysis. A Seurat object can contain separate RNA and SCT assays, but each downstream test must use the appropriate data layer intentionally.

Highly Variable Gene (HVG) selection (FindVariableFeatures()) identifies the top 2,000–5,000 genes with the highest cell-to-cell variation using the VST method. These HVGs carry the most

biological information; including all ~33,000 genes adds noise and increases computational cost. PCA reduces dimensionality from ~2,000–5,000 HVGs to 30–50 principal components. UMAP (Uniform Manifold Approximation and Projection) projects the PCA space into 2D for visualization 2. A critical caveat: UMAP is primarily a visualization. Local neighborhoods are often more reliable than long-range geometry; apparent distances, gaps, and cluster shapes can change with parameters. Biological conclusions should be checked in the higher-dimensional representation, marker genes, and replicate-level statistics.

1.3.4 Graph-Based Clustering and Cell Type Annotation

Seurat uses a **graph-based clustering** approach. FindNeighbors() builds a k-nearest neighbor (KNN) graph in PCA space (default k=20). FindClusters() applies the **Leiden algorithm** to detect communities in this graph. The resolution parameter controls granularity (higher = more clusters; 0.4–0.8 is typical). Cell type annotation uses three complementary approaches: (1) **manual annotation** using canonical marker genes (CD3E = T cells, CD68 = macrophages, EPCAM = epithelial cells); (2) **automated tools** (CellTypist, scType); and (3) **LLM-assisted annotation** — providing marker gene lists to GPT-4/Claude for annotation with reasoning.

1.3.5 Cell-Type Annotation Should Integrate Markers, References, Context, and AI

Cell-type annotation is an inference problem. A cluster label is strongest when independent evidence agrees: positive markers, absence of incompatible markers, tissue context, reference atlases, and an automated classifier. General-purpose LLMs are useful for generating candidate labels and explaining marker combinations because they can connect gene symbols to biomedical literature. They are weak, however, when a cluster is rare, transitional, disease-specific, or contaminated by ambient RNA. For these cases, the model may produce a fluent but biologically impossible label.

A practical evidence hierarchy is: (1) inspect the top marker genes and their effect sizes; (2) check negative markers that should not co-occur; (3) compare with a tissue-matched reference such as CELLxGENE/Human Cell Atlas; (4) run an independent annotation model such as CellTypist, scANVI, or a curated marker-scoring method; (5) ask an LLM or agent to generate competing labels with explicit supporting and contradictory evidence; and (6) retain an Unknown/Ambiguous label when the evidence does not converge.

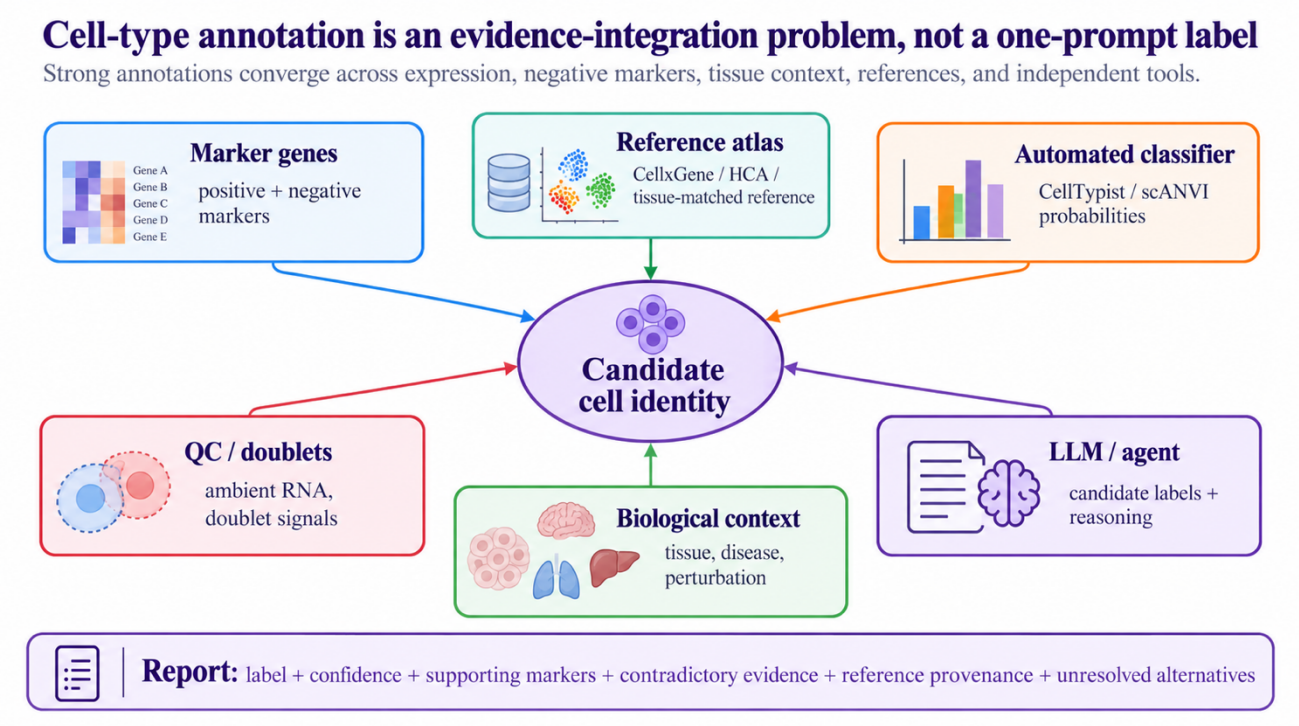

Figure 3. Evidence-aware cell-type annotation. The LLM is one evidence-generating component, not the final adjudicator.

Evidence layer	Example tools	Output	Verification issue
Marker evidence	FindAllMarkers / pseudobulk markers	Positive and negative markers; effect size	Strong for well-characterized cell types; vulnerable to ambient RNA.
Reference mapping	CELLxGENE, Azimuth, scANVI	Nearest reference labels / probabilities	Depends on tissue, disease, species, and atlas coverage.
Automated classifier	CellTypist / curated classifiers	Cell-type label + score	Fast and reproducible; may fail on unseen states.
LLM / agent	General LLM + retrieval	Candidate labels + rationale + citations	Useful for synthesis; must verify genes and sources.
Expert review	Domain knowledge + histology	Final label / ambiguity statement	Required for novel or transitional states.

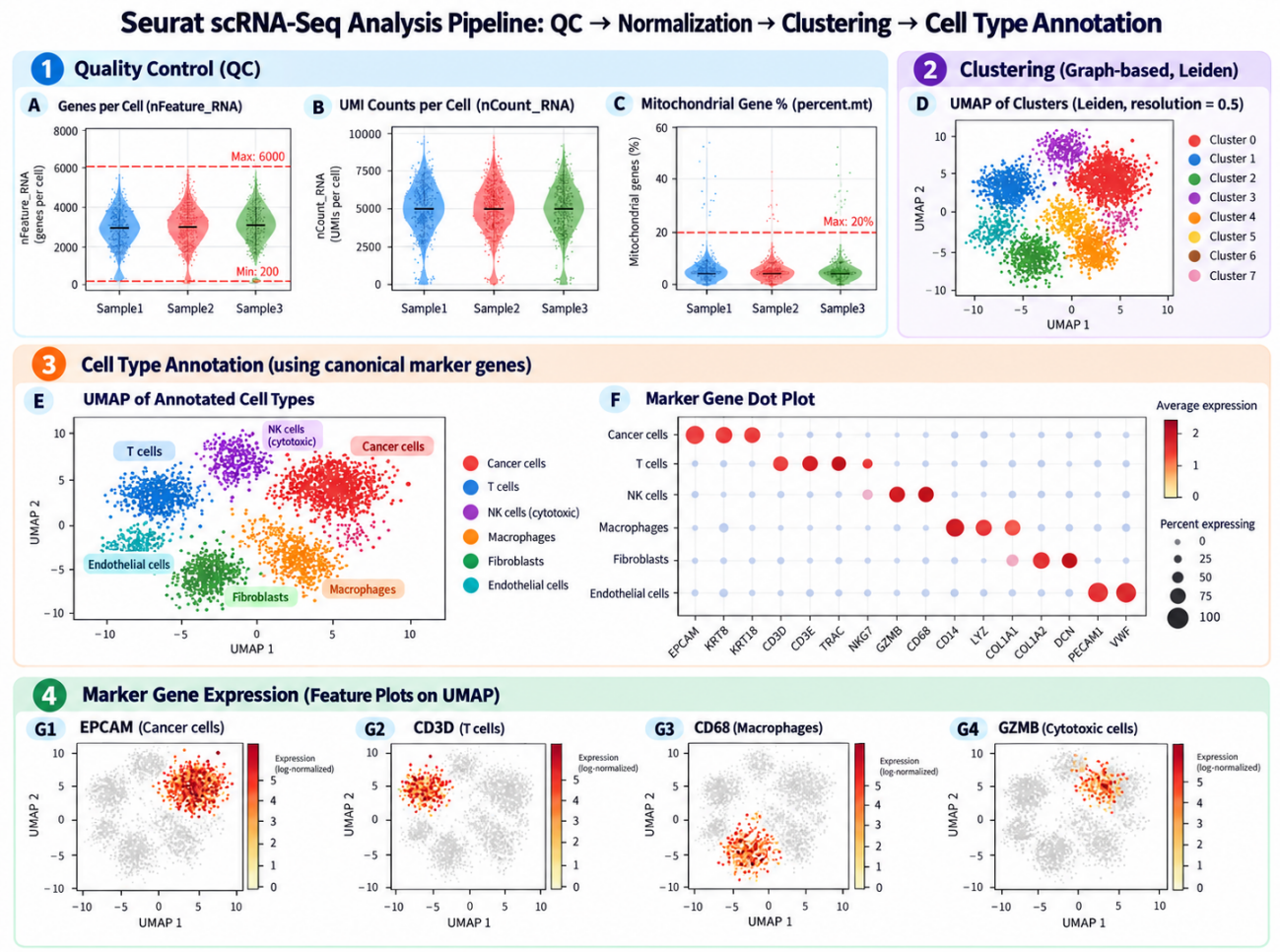

Figure 4. Seurat analysis pipeline outputs for a simulated tumor microenvironment dataset. (A–C) QC violin plots for nFeature_RNA, nCount_RNA, and percent.mt with filtering thresholds. (D) UMAP with Leiden clustering. (E) UMAP with annotated cell types and canonical marker genes. (F) Marker gene dot plot (size = % expressing; color = mean expression). (G1–G4) Feature plots for EPCAM, CD3E, CD68, and GZMB.

R Code Block 1 — Complete Seurat Workflow (10x Data → Annotated UMAP)

> ***LLM Prompt Template 1:*** *"I have a 10x Genomics scRNA-Seq dataset from a human tumor biopsy stored in 'filtered_feature_bc_matrix/' (~5,000 cells). Please write complete R code using Seurat v5 to: (1) load the 10x data and create a Seurat object; (2) calculate QC metrics and generate violin plots; (3) filter cells (nFeature_RNA 200–6000, percent.mt <20%); (4) normalize using SCTransform (vst.flavor='v2'); (5) run PCA (50 PCs) and generate an elbow plot; (6) run UMAP using top 30 PCs; (7) find neighbors (k=20) and cluster cells (resolution=0.5, Leiden algorithm); (8) find marker genes using FindAllMarkers (Wilcoxon, min.pct=0.25, logfc.threshold=0.25); (9) generate a UMAP colored by cluster, a dot plot of top 3 markers per cluster, and feature plots for EPCAM, CD3E, CD68, COL1A1; (10) save the Seurat object as 'tumor_seurat.rds'. Include detailed comments and print a summary of cells per cluster."*

```
library(Seurat); library(ggplot2); library(dplyr); library(patchwork)
```

```
# Step 1: Load 10x data
counts <- Read10X(data.dir = "filtered_feature_bc_matrix/")
seurat_obj <- CreateSeuratObject(counts=counts, project="TumorBiopsy",
                                 min.cells=3, min.features=200)
cat("Cells:", ncol(seurat_obj), "| Genes:", nrow(seurat_obj), "\n")

# Step 2: QC metrics
seurat_obj[["percent.mt"]] <- PercentageFeatureSet(seurat_obj, pattern="^MT-")
seurat_obj[["percent.rb"]] <- PercentageFeatureSet(seurat_obj, pattern="^RP[SL]")
VlnPlot(seurat_obj, features=c("nFeature_RNA","nCount_RNA","percent.mt"), ncol=3, pt.size=0.1)

# Step 3: Filter low-quality cells
seurat_filt <- subset(seurat_obj,
  subset=nFeature_RNA > 200 & nFeature_RNA < 6000 & percent.mt < 20)
cat("Cells after QC:", ncol(seurat_filt), "| Removed:", ncol(seurat_obj)-ncol(seurat_filt), "\n")

# Step 4: SCTransform normalization
seurat_filt <- SCTransform(seurat_filt, vars.to.regress="percent.mt",
                           vst.flavor="v2", verbose=FALSE)

# Step 5: PCA
seurat_filt <- RunPCA(seurat_filt, npcs=50, verbose=FALSE)
ElbowPlot(seurat_filt, ndims=50) + labs(title="PCA Elbow Plot (select 'elbow' — typically 20-35 PCs)")

# Step 6: UMAP
seurat_filt <- RunUMAP(seurat_filt, dims=1:30, verbose=FALSE)

# Step 7: Clustering (Leiden algorithm)
seurat_filt <- FindNeighbors(seurat_filt, dims=1:30, k.param=20, verbose=FALSE)
seurat_filt <- FindClusters(seurat_filt, resolution=0.5, algorithm=4, verbose=FALSE)
```

```
cat("\nCells per cluster:\n"); print(table(seurat_filt$seurat_clusters))
DimPlot(seurat_filt, reduction="umap", label=TRUE, label.size=4, pt.size=0.5) +
 labs(title="UMAP: Leiden Clustering (resolution=0.5)")

# Step 8: Marker genes
Idents(seurat_filt) <- "seurat_clusters"
markers <- FindAllMarkers(seurat_filt, only.pos=TRUE,
                          min.pct=0.25, logfc.threshold=0.25, test.use="wilcox")
top5 <- markers %>% group_by(cluster) %>% slice_max(order_by=avg_log2FC, n=5)
print(top5 %>% select(cluster, gene, avg_log2FC, pct.1, pct.2, p_val_adj))

# Step 9: Visualization
top3_genes <- markers %>% group_by(cluster) %>%
 slice_max(order_by=avg_log2FC, n=3) %>% pull(gene) %>% unique()
DotPlot(seurat_filt, features=top3_genes) + RotatedAxis() +
 labs(title="Top 3 Marker Genes per Cluster")
FeaturePlot(seurat_filt, features=c("EPCAM","CD3E","CD68","COL1A1"),
            ncol=2, pt.size=0.3, order=TRUE, cols=c("lightgrey","#E53935"))

# Step 10: Manual annotation
cluster_annotations <- c("0"="Cancer cells","1"="CD8+ T cells","2"="Macrophages",
                         "3"="Fibroblasts","4"="Endothelial","5"="B cells",
                         "6"="NK cells","7"="Mast cells")
seurat_filt[["cell_type"]] <- cluster_annotations[as.character(seurat_filt$seurat_clusters)]
DimPlot(seurat_filt, reduction="umap", group.by="cell_type",
        label=TRUE, label.size=3.5, pt.size=0.5) + labs(title="UMAP: Annotated Cell Types")
saveRDS(seurat_filt, "tumor_seurat.rds")
cat("Saved: tumor_seurat.rds |", ncol(seurat_filt), "cells |",
    length(unique(seurat_filt$seurat_clusters)), "clusters\n")
```

1.4 Trajectory Analysis and RNA Velocity

Pseudotime Analysis with Monocle3. Pseudotime assigns each cell a value representing its position along a developmental trajectory, from the earliest state (e.g., a stem cell) to the latest state (e.g., a terminally differentiated cell). **Monocle3** constructs a **principal graph** through the UMAP embedding and orders cells along its branches 4. The user specifies one or more "root" cells (typically the most undifferentiated state), and Monocle3 calculates pseudotime as the geodesic distance from the root. Key applications include: reconstructing stem cell differentiation hierarchies; identifying the order of gene expression changes during development; detecting branching points where cells commit to different lineages.

RNA Velocity with scVelo. RNA velocity predicts the **future** transcriptional state of each cell based on the ratio of unspliced (nascent pre-mRNA) to spliced (mature mRNA) transcript counts 5. The biological basis: a gene being actively transcribed will have more unspliced transcripts relative to its steady-state spliced level; a gene being downregulated will have fewer. **scVelo** implements a dynamical model estimating splicing kinetics (transcription rate, splicing rate, degradation rate) for each gene, providing more accurate velocity estimates than the original steady-state model. RNA velocity is visualized as arrows on the UMAP plot — arrow direction = predicted future state; arrow length = velocity magnitude.

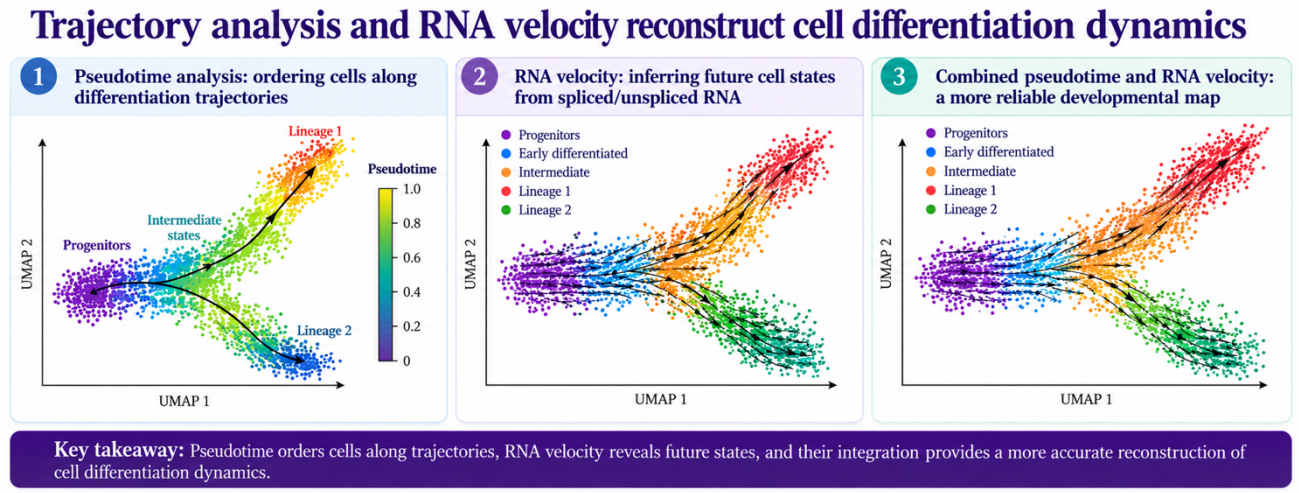

Figure 5. Three views of the same hematopoiesis dataset. (A) Pseudotime analysis (Monocle3): cells ordered from HSC (pseudotime=0, dark purple) through progenitors to terminally differentiated monocytes, granulocytes, T cells, and B cells (pseudotime=1, yellow). (B) Cell type annotation: same UMAP colored by cell type identity. (C) RNA velocity (scVelo): arrows show predicted differentiation directions, confirming the pseudotime trajectories and revealing the myeloid/lymphoid branching point.

The key difference between pseudotime and RNA velocity is their information source: pseudotime uses the **current** transcriptional state to infer relative developmental order; RNA velocity uses **transcriptional dynamics** (unspliced/spliced ratios) to predict the **future** state. RNA velocity requires re-processing raw FASTQ files with **velocyto** or **STARsolo** to generate spliced/unspliced count matrices.

SESSION 2 (40 min): Spatial Transcriptomics — Mapping Gene Expression in Tissue Space

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2.1 Why Spatial Resolution? The Missing Dimension of scRNA-Seq

Single-cell RNA sequencing resolves cellular heterogeneity but requires tissue dissociation — destroying the spatial organization of the tissue. Three specific biological phenomena illustrate why spatial information is scientifically indispensable.

Problem 1: Tissue Architecture and Metabolic Zonation. The liver is organized into functional lobules with a central vein at the center and portal triads at the periphery. Hepatocytes in different zones perform fundamentally different metabolic functions: pericentral hepatocytes specialize in lipid metabolism and drug detoxification; periportal hepatocytes specialize in gluconeogenesis and urea synthesis. This **metabolic zonation** depends entirely on the hepatocyte's position relative to oxygen and nutrient gradients. scRNA-Seq can identify that two hepatocyte populations exist, but cannot reveal their spatial relationship or the gradient nature of the zonation. Spatial transcriptomics maps this gradient directly onto the tissue section 6.

Problem 2: Cell-Cell Communication in Spatial Context. Ligand-receptor interactions between adjacent cells are fundamentally spatial events. A cancer cell secreting TGF-β affects immediately adjacent fibroblasts, not fibroblasts on the other side of the tumor. Spatial transcriptomics provides the physical proximity information necessary to validate and prioritize these paracrine interactions.

Problem 3: Spatially Restricted Cell Populations. Some cell types are defined by their spatial position rather than their transcriptome alone — Cajal-Retzius cells in the brain cortex, Paneth cells at the base of intestinal crypts, and pericentral hepatocytes in the liver lobule. Without spatial information, these populations may be misclassified or their functional significance misunderstood.

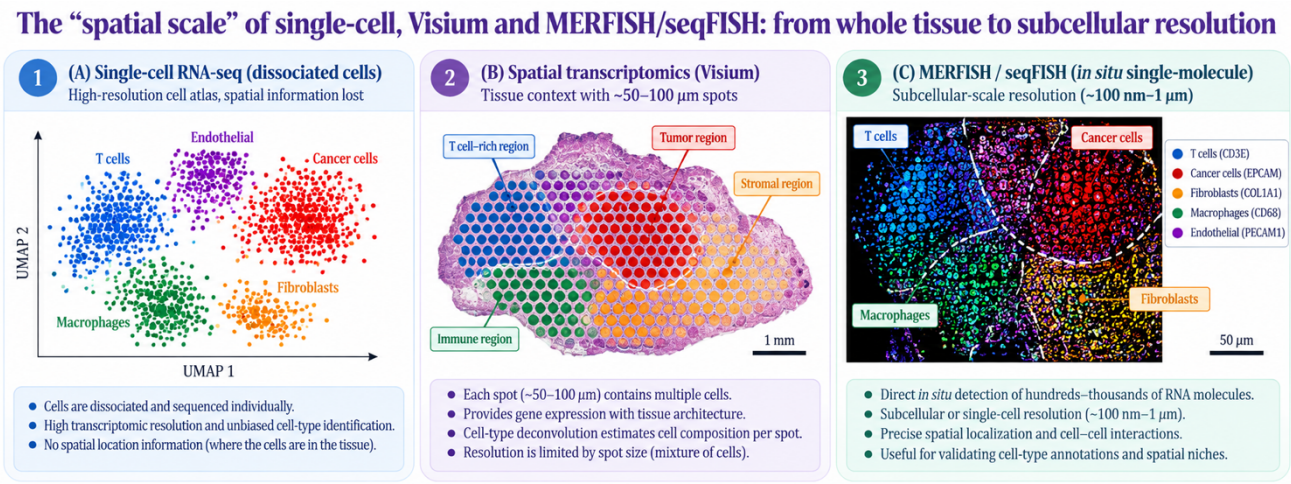

Figure 6. The spatial dimension illustrated with liver metabolic zonation. (A) scRNA-Seq UMAP: Zone 1 and Zone 2 hepatocytes cluster separately but their spatial relationship is unknown. (B) Liver lobule anatomy: hexagonal structure with central vein (CV) and portal triads (PT), showing three metabolic zones. (C) 10x Visium spatial transcriptomics: metabolic zonation directly visualized, with Zone 1 (pericentral, red), Zone 2 (mid-lobular, blue), and Zone 3 (periportal, green) hepatocytes mapped to their anatomical positions.

2.2 Spatial Transcriptomics Technologies: Three Generations

Generation 1: Spot-Based Array Capture. Classic 10x Genomics Visium uses barcoded capture spots (55 µm diameter) and is still widely used in archived and published datasets. Visium HD now provides a continuous 2 µm barcoded grid that can be analyzed as fine bins or with image-based cell segmentation, substantially changing the analysis workflow. mRNA diffuses into spots, is captured by poly-dT oligonucleotides, and sequenced. Each spot captures mRNA from approximately 1–10 cells. 10x Visium HD (2024) uses 2 µm × 2 µm bins, achieving effective single-cell resolution. Stereo-seq (BGI/MGI) uses a DNB array with 500 nm spot size — the highest resolution of any sequencing-based spatial platform, developed by BGI's MGI Tech subsidiary.

Generation 2: In-Situ Sequencing and Imaging-Based Methods. MERFISH and seqFISH+ measure a predefined panel of genes (100–10,000) directly in intact tissue sections using combinatorial FISH barcoding, achieving true single-cell or subcellular resolution. Xenium (10x Genomics) is a targeted in situ platform with subcellular transcript localization. Current Xenium Prime chemistry supports 5,000-gene panels (plus custom add-ons), while focused panels remain available for smaller targeted studies.

Generation 3: Multimodal Spatial Omics. Spatial CITE-seq measures mRNA and surface proteins with spatial coordinates. **Spatial ATAC-seq** maps chromatin accessibility in spatial context. **CODEX/PhenoCycler** (Akoya Biosciences) measures >50 proteins simultaneously with spatial coordinates — effectively a spatial proteomics platform.

Technology	Resolution	Gene Coverage	Principle	Cost/Sample	Key Software	Best Application
10x Visium	55 µm (1–10 cells/spot)	~20,000	Sequencing	~$500–800	Seurat, Squidpy	Clinical samples; FFPE compatible
10x Visium HD	2 µm (single cell)	~20,000	Sequencing	~$800–1,200	Seurat, Squidpy	High-resolution tissue mapping
Stereo-seq (BGI)	500 nm (subcellular)	~20,000	Sequencing	~$300–600	SAW, Seurat	Ultra-high resolution; atlas projects
MERFISH	Single cell / subcellular	100–10,000 (panel)	Imaging (FISH)	~$1,000–3,000	Vizgen MERSCOPE	Brain mapping; cell-cell communication
seqFISH+	Single cell	~10,000 (panel)	Imaging (FISH)	~$2,000–5,000	Custom	Isoform resolution; subcellular localization
Xenium (10x)	Subcellular	Up to 5,000 genes (Prime); focused panels also available	In situ imaging / decoding	~$500–1,000	Xenium Explorer	Clinical diagnostics; tumor microenvironment
SLIDE-seq v2	10 µm	~20,000	Sequencing	~$200–400	Seurat, Squidpy	Brain; rapid tissue mapping

Table 4. Spatial transcriptomics technology comparison: resolution, gene coverage, measurement principle, cost, software, and best application [6] [7].

2.3 Spatial Transcriptomics Analysis Pipeline

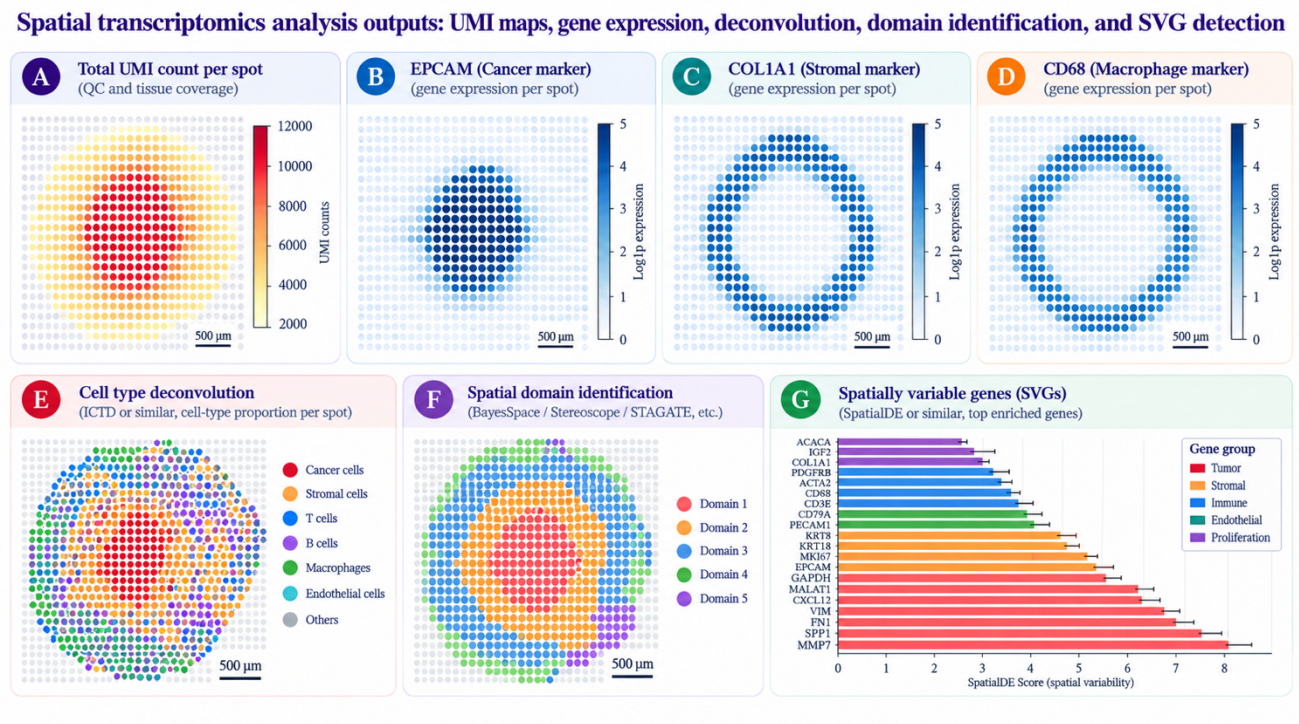

Figure 7. Spatial transcriptomics analysis outputs for a simulated breast cancer tissue section. (A) Total UMI count per spot (QC). (B–D) Spatial expression maps for EPCAM (tumor), COL1A1 (stroma), and CD68 (macrophage). (E) Cell type deconvolution (RCTD): dominant cell type per spot. (F) Spatial domain identification (BayesSpace): tumor core, tumor microenvironment, and normal tissue. (G) Spatially variable genes (SVGs, SpatialDE): top 20 spatially structured genes grouped by cell type.

Spot Deconvolution. Because each Visium spot contains multiple cells, **deconvolution** estimates cell type proportions per spot using a paired scRNA-Seq reference. **RCTD** (Robust Cell Type Decomposition) uses a statistical model accounting for platform differences. **cell2location** is a Bayesian model providing more accurate estimates for rare cell types. **Tangram** uses deep learning to map scRNA-Seq cell types onto spatial coordinates 7.

Spatially Variable Gene (SVG) Detection. SVGs are genes with spatially structured expression patterns. **SpatialDE** uses a Gaussian process model. **SPARK-X** is a faster non-parametric alternative. **Moran's I** statistic measures spatial autocorrelation — high Moran's I indicates nearby spots tend to have similar expression (spatially structured).

Spatial Domain Identification. BayesSpace uses a Bayesian model with a Markov random field prior encouraging spatially adjacent spots to be assigned to the same domain. **BANKSY** uses neighborhood expression profiles as additional clustering features. **GraphST** uses a graph neural network learning from both gene expression and spatial graphs simultaneously.

2.3.1 Cutting-Edge Spatial AI: From Coordinates to Niche and Histology-Aware Models

Spatial analysis increasingly uses models that learn from cell neighborhoods rather than treating coordinates as a plotting layer. Nicheformer integrates dissociated and spatial transcriptomes and can transfer spatial context back onto non-spatial scRNA-seq cells. Novae is a graph-based foundation model that learns hierarchical spatial domains and supports zero-shot domain inference across

multiple panels and technologies. These methods are useful when the scientific question concerns niches, boundaries, or tissue organization rather than only cell identity.

A second frontier links histology to transcriptomics. OmiCLIP/Loki (Nature Methods, 2025) was trained on millions of paired H&E image patches and spatial-transcriptomic measurements. It supports cross-modal retrieval, annotation, decomposition, and prediction of spatial gene expression from tissue morphology. Such models can reduce experimental cost or guide region selection, but predicted expression from an H&E image is not a substitute for measured RNA. The predicted map should be labeled as inferred and validated in an independent measured dataset.

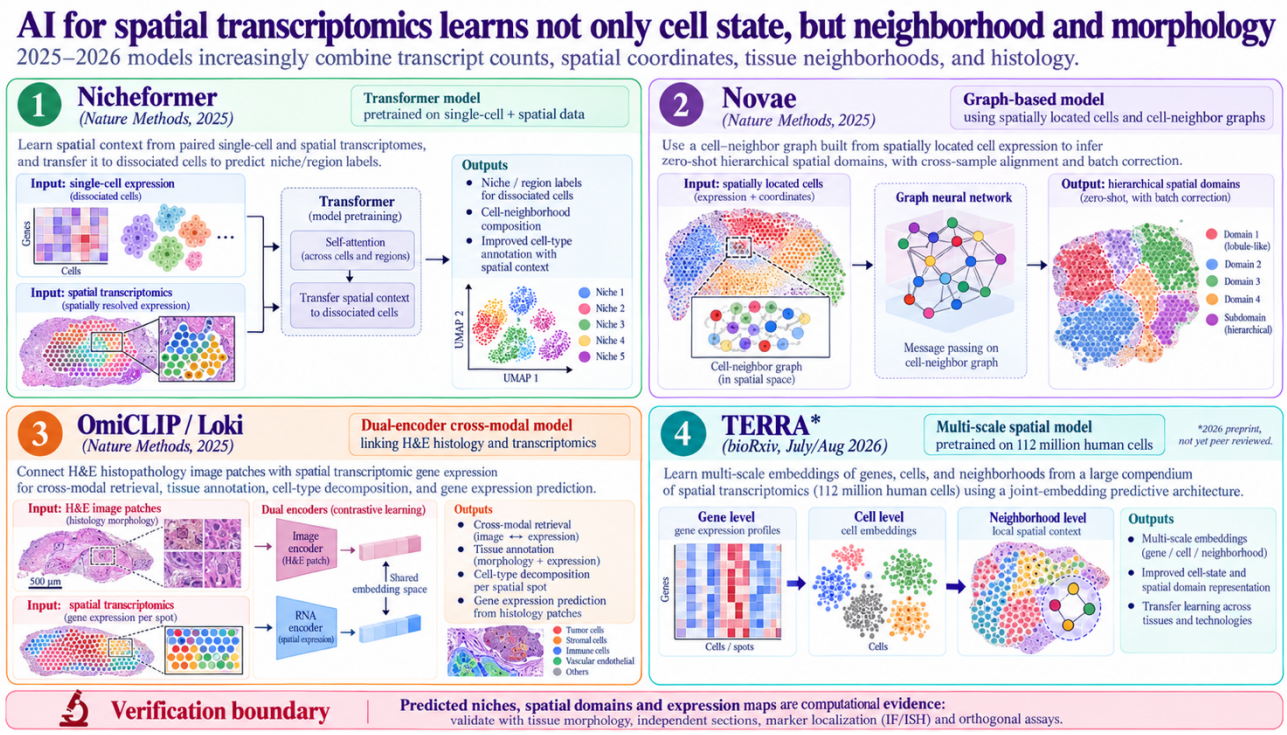

Figure 8. Spatial AI landscape. Modern models learn cell state, neighborhood, and morphology, but their outputs remain computational evidence requiring tissue-level validation.

Cell-Cell Communication Analysis. CellChat and **NicheNet** infer ligand-receptor interactions between cell types. **COMMOT** (Communicating Objects via Manifold Transport) uses optimal transport theory to infer spatially resolved cell-cell communication, explicitly accounting for physical distance between cells.

2.3.2 Worked AI-Agent Case: From a Cell State to a Spatially Testable Mechanism

Consider a liver study in which scRNA-seq suggests expansion of an SPP1/TREM2-high macrophage population. An agent can help by: checking that the macrophage expansion occurs across biological replicates rather than one donor; generating pseudobulk contrasts for macrophages; comparing marker genes with a reference atlas; transferring the annotation to a spatial dataset; quantifying whether SPP1-high macrophages are enriched near activated stellate cells; and drafting a candidate ligand-receptor or niche hypothesis.

The agent must not collapse distinct evidence levels. “SPP1 and COL1A1 are spatially close” is a colocalization statement. “Macrophages signal to stellate cells through SPP1” is a communication hypothesis requiring receptor expression and directional evidence. “SPP1 drives fibrosis” is a causal claim requiring perturbation or another causal design. The agent should preserve these verbs and explicitly state which transition in the causal chain remains untested.

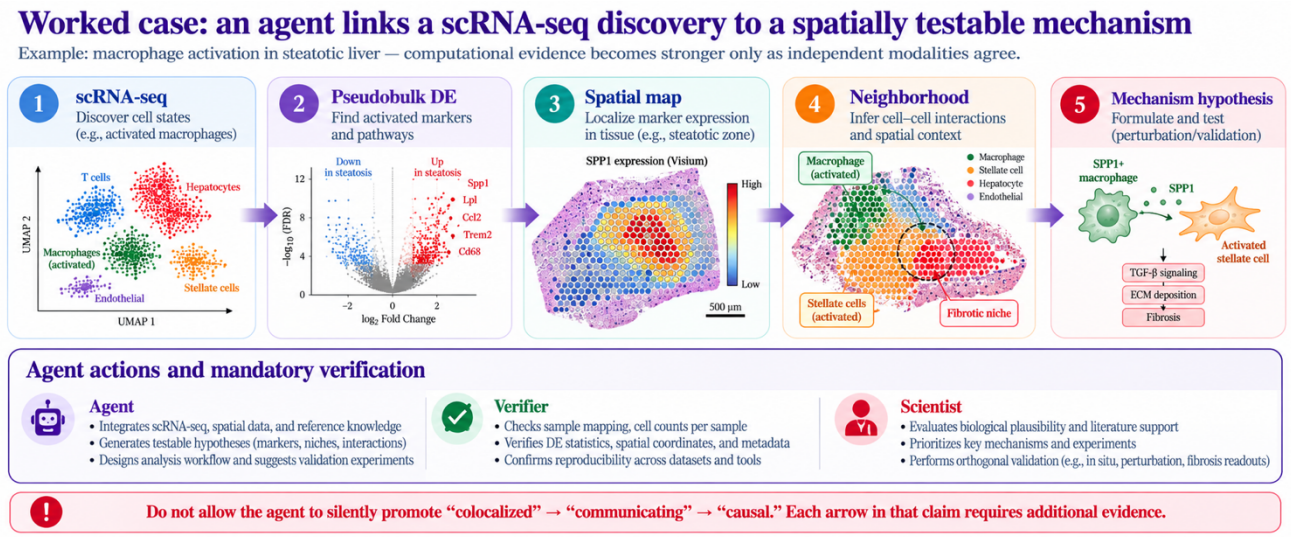

Figure 9. Worked example showing how an agent can connect single-cell and spatial evidence without overstating causality.

R Code Block 2 — Seurat Spatial Analysis Pipeline (10x Visium → Deconvolution → SVG Detection)

> ***LLM Prompt Template 2:*** *"I have a 10x Visium spatial transcriptomics dataset from a human breast cancer tissue section in 'visium_output/' (contains: filtered_feature_bc_matrix/, spatial/). I also have a paired scRNA-Seq Seurat object 'tumor_seurat.rds' with cell type annotations in the 'cell_type' metadata column. Please write complete R code using Seurat v5 to: (1) load the Visium data; (2) perform QC (plot UMI counts and gene counts per spot on the tissue); (3) normalize using SCTransform; (4) run PCA, UMAP, and clustering; (5) find spatially variable features (method='markvariogram', nfeatures=3000); (6) generate spatial feature plots for EPCAM, COL1A1, CD68, CD3E; (7) generate a spatial dimension plot colored by cluster; (8) perform RCTD cell type deconvolution using the paired scRNA-Seq reference; (9) plot cell type proportions on the tissue. Include detailed comments."*

```
library(Seurat); library(ggplot2); library(dplyr); library(patchwork)
```

```
# Step 1: Load 10x Visium data
spatial_obj <- Load10X_Spatial(data.dir="visium_output/",
  filename="filtered_feature_bc_matrix.h5", assay="Spatial",
  slice="breast_cancer", filter.matrix=TRUE)
cat("Spots:", ncol(spatial_obj), "| Genes:", nrow(spatial_obj), "\n")

# Step 2: QC — visualize on tissue image
SpatialFeaturePlot(spatial_obj, features="nCount_Spatial",
  pt.size.factor=1.6, alpha=c(0.1,1)) + labs(title="Total UMI Count per Spot")
spatial_obj <- subset(spatial_obj,
  subset=nCount_Spatial > 500 & nFeature_Spatial > 200)
cat("Spots after QC:", ncol(spatial_obj), "\n")
```

```r
# Step 3: SCTransform normalization
spatial_obj <- SCTransform(spatial_obj, assay="Spatial",
 vars.to.regress="nCount_Spatial", vst.flavor="v2", verbose=FALSE)

# Step 4: Dimensionality reduction and clustering
spatial_obj <- RunPCA(spatial_obj, assay="SCT", verbose=FALSE)
spatial_obj <- FindNeighbors(spatial_obj, reduction="pca", dims=1:30, verbose=FALSE)
spatial_obj <- FindClusters(spatial_obj, verbose=FALSE, resolution=0.5)
spatial_obj <- RunUMAP(spatial_obj, reduction="pca", dims=1:30, verbose=FALSE)
SpatialDimPlot(spatial_obj, label=TRUE, label.size=3, pt.size.factor=1.6) +
 labs(title="Spatial Clustering (Leiden clusters on H&E image)")

# Step 5: Spatially Variable Features (SVGs)
spatial_obj <- FindSpatiallyVariableFeatures(spatial_obj, assay="SCT",
 features=VariableFeatures(spatial_obj)[1:3000],
 selection.method="markvariogram", nfeatures=1000)
top_svgs <- SpatiallyVariableFeatures(spatial_obj,
 selection.method="markvariogram")[1:20]
cat("Top 20 SVGs:\n"); print(top_svgs)
SpatialFeaturePlot(spatial_obj, features=top_svgs[1:6],
 ncol=3, alpha=c(0.1,1), pt.size.factor=1.4) +
 plot_annotation(title="Top 6 Spatially Variable Genes")

# Step 6: Canonical marker expression on tissue
SpatialFeaturePlot(spatial_obj, features=c("EPCAM","COL1A1","CD68","CD3E"),
 ncol=2, alpha=c(0.1,1), pt.size.factor=1.4) +
 plot_annotation(title="Canonical Cell Type Markers — Spatial Expression")

# Step 7: RCTD cell type deconvolution
library(spacexr)
sc_ref <- readRDS("tumor_seurat.rds")
reference <- Reference(
 counts=GetAssayData(sc_ref, assay="RNA", slot="counts"),
 cell_types=factor(sc_ref$cell_type), nUMI=sc_ref$nCount_RNA)
counts_spatial <- GetAssayData(spatial_obj, assay="Spatial", slot="counts")
coords <- GetTissueCoordinates(spatial_obj)
query <- SpatialRNA(coords, counts_spatial, colSums(counts_spatial))
RCTD_obj <- create.RCTD(query, reference, max_cores=4)
RCTD_obj <- run.RCTD(RCTD_obj, doublet_mode="full")
results_df <- as.data.frame(RCTD_obj@results$weights)
results_df <- results_df / rowSums(results_df)
for (ct in colnames(results_df)) {
```

```
  spatial_obj[[paste0("prop_",gsub(" ","_",ct))]] <- results_df[,ct]
}
SpatialFeaturePlot(spatial_obj,
  features=paste0("prop_",gsub(" ","_",c("Cancer cells","Macrophages","CD8+ T cells","Fibroblasts"))),
  ncol=2, alpha=c(0.1,1), pt.size.factor=1.4) +
  plot_annotation(title="Cell Type Proportions per Spot (RCTD Deconvolution)")
```

```
saveRDS(spatial_obj, "visium_spatial_seurat.rds")
```

2.4 AI and LLM Agent Integration in Single-Cell Analysis

The integration of artificial intelligence into single-cell and spatial transcriptomics analysis operates at three distinct tiers of sophistication.

2.4.1 From "LLM Writes Code" to an Inspectable AI Agent

An AI assistant becomes an agent when it can select tools, execute code, inspect outputs, retain state, and decide the next action. In single-cell analysis, this is powerful because a real workflow is inherently iterative: the analyst inspects QC, changes thresholds, reruns normalization or integration, examines markers, revises annotations, and only then performs replicate-aware differential testing. An agent can automate this loop, but only if each action is observable and reversible.

For teaching, use a six-stage contract: Question + design -> Agent plan -> Tool execution -> Observed outputs -> Verification -> Human decision. The agent should never silently change sample labels, reference genome, filtering thresholds, or statistical contrasts. Every run should emit a manifest containing input files, sample-to-condition mapping, object dimensions before and after filtering, software/package versions, random seeds, warnings, and saved outputs.

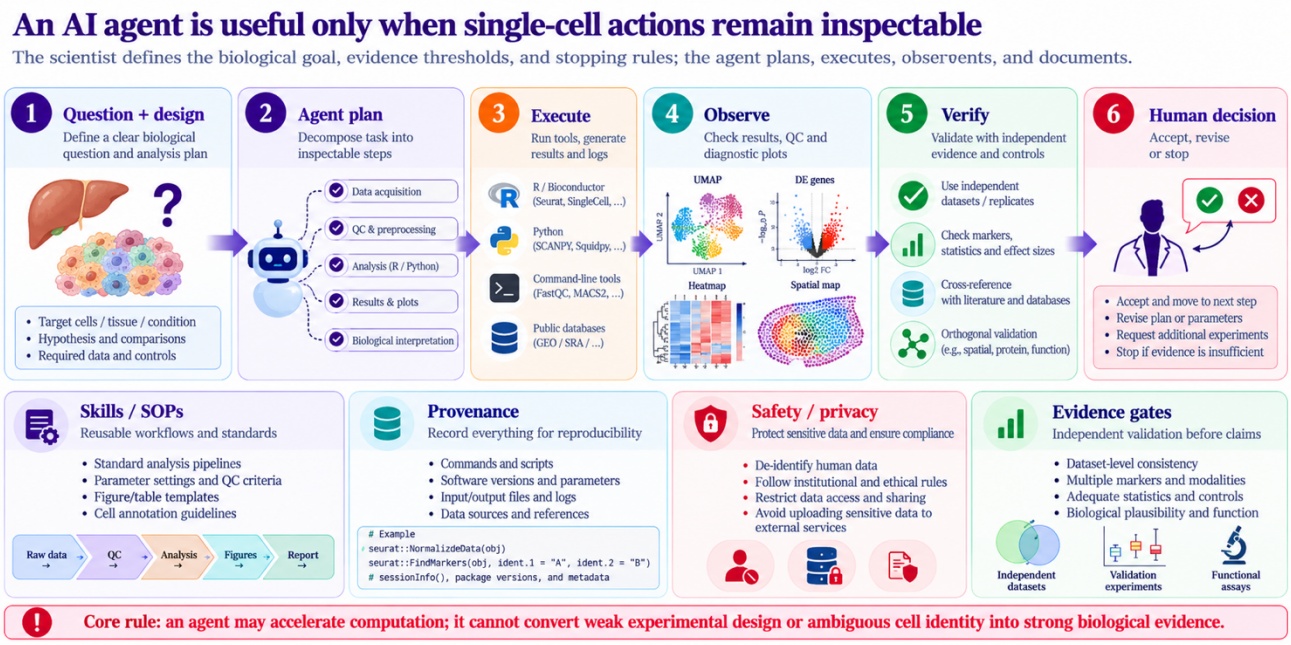

Figure 10. Inspectable AI-agent workflow for single-cell analysis. The scientist owns the question, constraints, verification gates, and final biological interpretation.

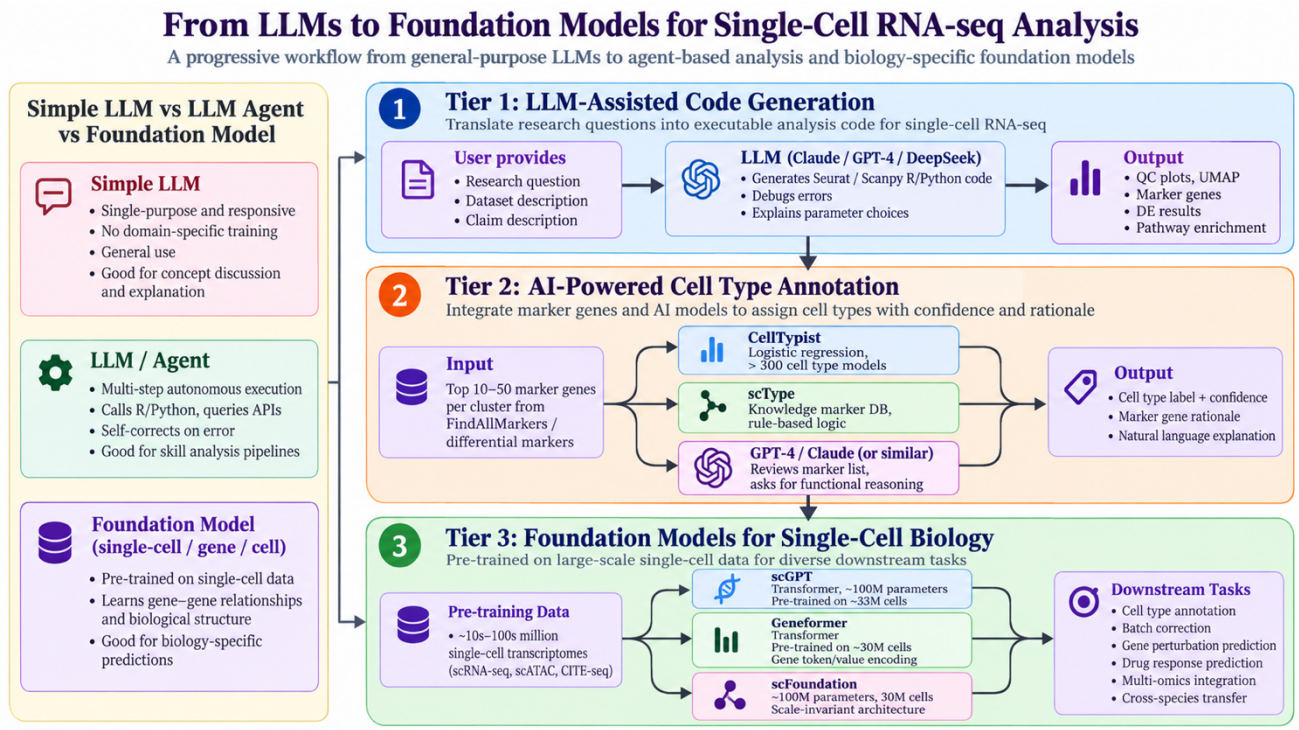

Figure 11. The three-tier AI integration landscape for single-cell analysis. Tier 1 (blue): LLM-assisted code generation using GPT-4, Claude, or DeepSeek. Tier 2 (orange): AI-powered cell type annotation using CellTypist, scType, and LLM-assisted marker interpretation. Tier 3 (green): Foundation models (scGPT, Geneformer, scFoundation) pre-trained on tens of millions of single-cell transcriptomes, enabling biology-specific predictions including perturbation response, batch correction, and drug response prediction.

Tier 1 — LLM-Assisted Code Generation. The most immediately practical application of LLMs is code generation and debugging. LLMs can write complete Seurat or Scanpy pipelines from natural language descriptions, explain parameter choices, debug error messages, and suggest alternatives. Code Blocks 1–4 in this handout were generated using LLM prompt templates — demonstrating that students can produce publication-quality analysis code without memorizing syntax.

2.4.3 Recent Agentic Systems: Standardized Ingestion, Trustworthy Annotation, and Interactive Spatial Analysis

Three recent systems illustrate how agentic bioinformatics differs from prompt-only assistance. First, an npj Artificial Intelligence study published in January 2026 described an agentic framework for ingesting and standardizing public scRNA-seq datasets: the agent extracts metadata from papers, retrieves repository data, and executes a standardized analysis while maintaining structured outputs. Second, CellTypeAgent combines LLM-proposed cell identities with external database verification rather than accepting the LLM label directly. Third, STAT (2026 preprint) uses multiple agents and a persistent spatial-analysis session so that users can iteratively inspect tissue views and intermediate results instead of delegating the entire analysis as a black box.

For students, these systems motivate a practical design rule: use separate agent roles for data stewardship, QC, annotation, statistics, spatial analysis, and verification. A supervisor agent can coordinate the sequence, but a verifier should be logically independent of the agent that generated the claim. This separation reduces confirmation bias and makes it easier to identify whether an error originated in metadata, code, annotation, or interpretation.

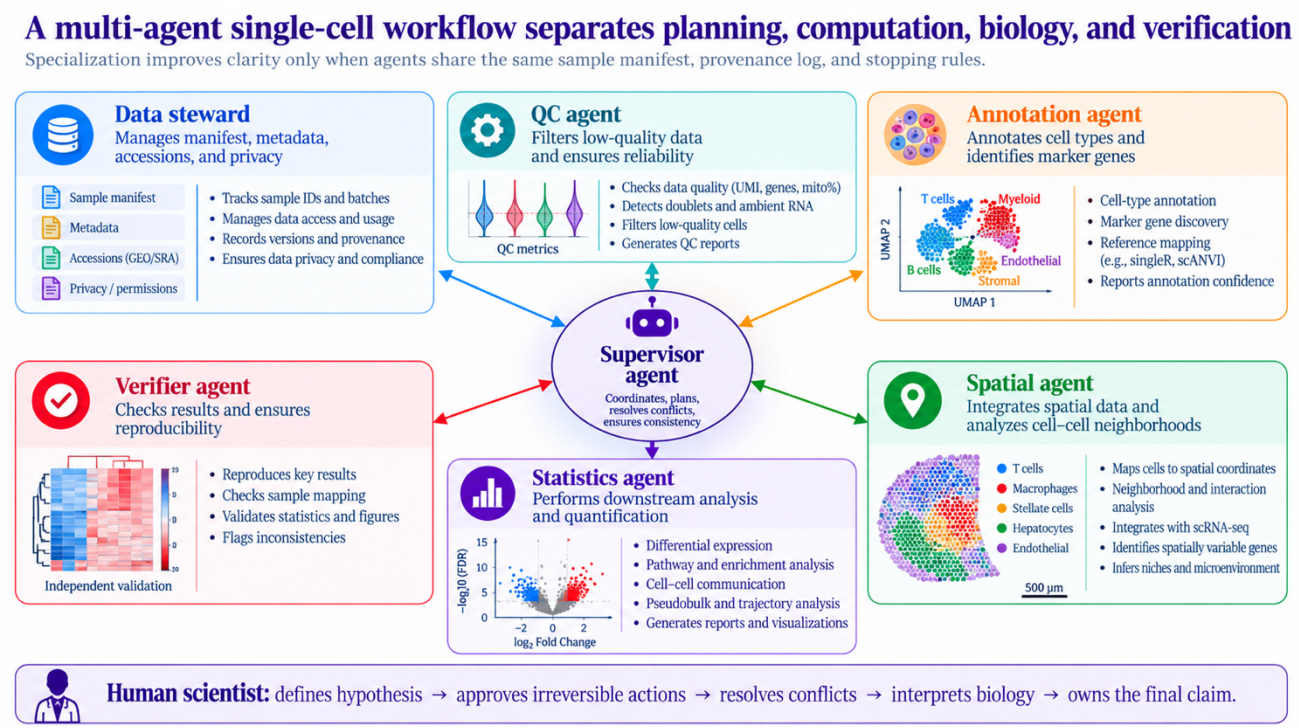

Figure 12. Multi-agent architecture for single-cell and spatial analysis. Shared metadata and provenance are more important than the number of agents.

Tier 2 — AI-Powered Cell Type Annotation. CellTypist is a logistic regression classifier trained on >400 human and mouse cell type models from 20+ tissues. **scType** uses a curated marker gene database (ScTypeDB) and scores each cluster based on positive and negative marker enrichment — runs entirely in R without a pre-trained model. **GPT-4/Claude for marker gene interpretation** leverages the LLM's training on the biomedical literature to recognize cell type signatures not captured in any single reference database. However, LLM annotations must always be validated — LLMs can produce plausible-sounding but incorrect annotations for rare or poorly characterized cell types.

Tier 3 — Foundation Models for Single-Cell Biology. scGPT (2024) is a transformer-based foundation model pre-trained on >33 million human single-cell transcriptomes from the CellxGene database 8. Like GPT for text, scGPT processes **gene tokens** — each gene is a token, and the model learns contextual gene-gene relationships across millions of cells. scGPT can be fine-tuned for cell type annotation, batch correction, gene perturbation prediction, and multi-omics integration. **Geneformer** (2023) is pre-trained on 30 million single-cell transcriptomes using **gene rank-value encoding** — genes are ranked by expression level within each cell, making the representation robust to technical variation 9. Geneformer demonstrated the ability to predict gene knockout effects in silico, identifying novel therapeutic targets for cardiomyopathy by simulating the transcriptional effects of silencing each gene in cardiomyocytes. **scFoundation** (2024) is a 100-million-parameter model pre-trained on >50 million cells, achieving state-of-the-art performance on multiple benchmarks.

2.4.2 2025-2026 Update: Foundation Models Are Expanding - and Being Benchmarked More Critically

The frontier has moved from one “GPT for cells” model to a diverse ecosystem of models optimized for different tasks. scPRINT (Nature Communications, 2025) was pretrained on 50 million cells and focuses on cell-specific gene-network inference. Nicheformer (Nature Methods, 2025) was trained on a corpus of more than 110 million dissociated and spatially resolved cells and explicitly learns spatial context. Novae (Nature Methods, 2025) uses graph-based pretraining across nearly 30 million

spatially resolved cells to support zero-shot spatial-domain inference across tissues and technologies. These models illustrate an important shift: the representation can encode not only cell state but also neighborhood and tissue organization.

At the same time, independent benchmarking has challenged the assumption that a larger pretrained model automatically produces a better biological representation. A 2025 Nature Methods evaluation found that scGPT and Geneformer embeddings did not consistently outperform simpler baselines for several zero-shot tasks. This is an essential lesson for students: foundation-model output should be compared with transparent baselines such as PCA/scVI, validated on held-out datasets, and judged by biological utility rather than model size or novelty.

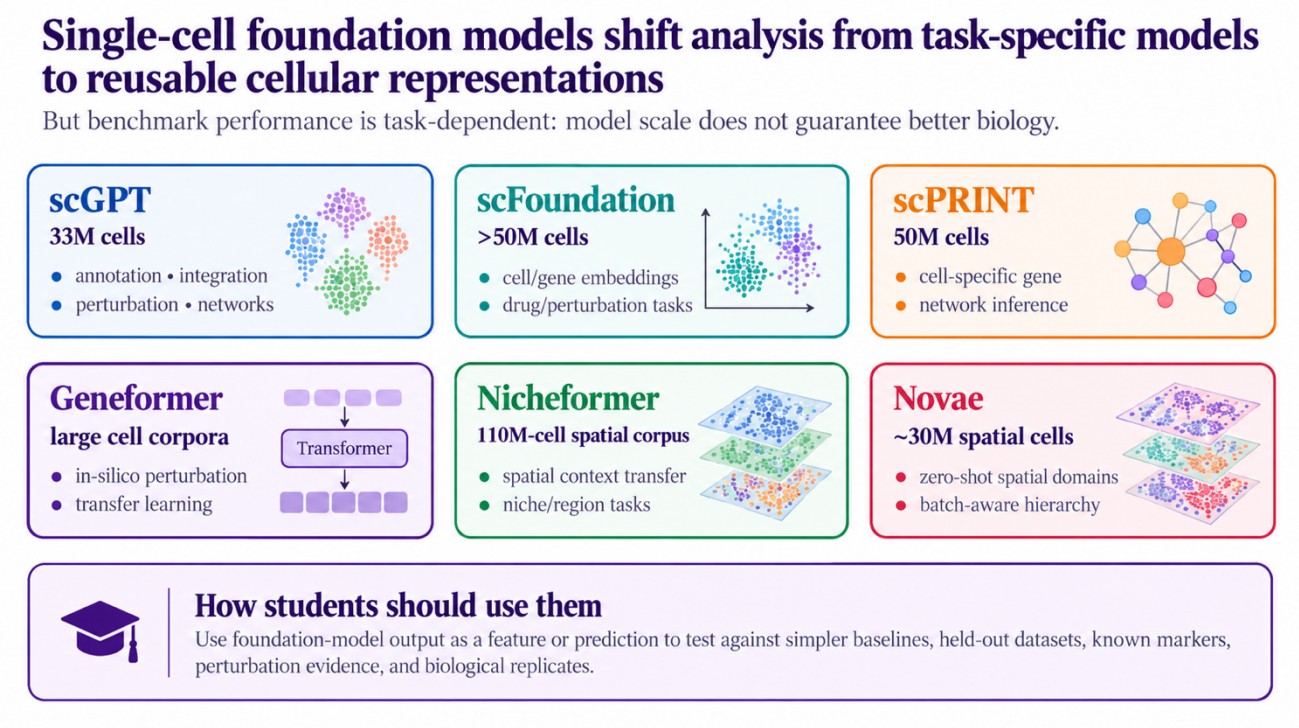

Figure 13. Foundation-model landscape for single-cell and spatial omics. Different models encode different inductive biases and should be evaluated task by task.

Model	Peer-reviewed	Pretraining scale	Representative tasks	Interpretation rule
scGPT	2024	>33M cells	Cell annotation, integration, perturbation, gene networks	Use as pretrained representation; verify against simpler baselines.
scFoundation	2024	>50M cells	Expression enhancement, perturbation/drug response, annotation	Large model does not eliminate batch/design constraints.
scPRINT	2025	50M cells	Cell-specific gene-network inference	Network edges are predictions, not causal interactions.
Nicheformer	2025	SpatialCorpus-110M	Spatial context transfer, niche/region tasks	Especially relevant when spatial context is the target.
Novae	2025	~30M spatial cells	Zero-shot spatial domains, batch-aware hierarchy	Domain boundaries still require morphology and marker validation.

AI Tool	Training Data	Architecture	Key Tasks	Input	Output	Access
scGPT	33M human cells (CellxGene)	Transformer (GPT-style)	Annotation, batch correction, perturbation prediction	Gene expression matrix	Cell embeddings, predictions	GitHub + HuggingFace
Geneformer	30M cells	Transformer (BERT-style)	Perturbation prediction, network inference	Gene rank-value encoding	Cell embeddings, gene importance	HuggingFace
scFoundation	50M cells	Transformer (100M params)	Annotation, imputation, integration	Gene expression matrix	Cell embeddings, predictions	GitHub
UCE	36M cells (multi-species)	Transformer	Cross-species transfer, zero-shot annotation	Gene expression matrix	Universal cell embeddings	GitHub
CellTypist	>400 cell type models	Logistic regression	Cell type annotation	Normalized expression matrix	Cell type labels + probabilities	PyPI + R
scType	Curated marker DB	Marker gene scoring	Cell type annotation	Seurat object	Cell type labels	GitHub (R)
GPT-4/Claude	General LLM (text)	Transformer (LLM)	Code generation, marker interpretation, results writing	Natural language + marker genes	Code, annotations, text	API / Web

Table 5. AI and foundation model tools for single-cell analysis: training data, architecture, key tasks, input, output, and access [8] [9].

R Code Block 3 — LLM-Assisted Cell Type Annotation (scType + GPT-4 Prompt)

CARE+V Prompt Template for a Single-Cell / Spatial Agent

Context: Human liver scRNA-seq and matched spatial transcriptomics, with sample_id, condition, donor, cell_type, and tissue coordinates. Action: inspect metadata, propose a QC/annotation plan, execute only reversible analysis steps, produce one pseudobulk contrast and one spatial validation figure. Requirements: use raw counts for aggregation; never treat cells as independent biological replicates; retain donor identity; report object dimensions before/after each filter; save code, logs, sessionInfo(), and all intermediate tables. Examples: provide the expected columns and one example marker set. Verification (+V): list assumptions, potential doublets/ambient RNA, alternative cell labels, coefficient direction, references/databases checked, and one independent test that could falsify the final interpretation.

> ***LLM Prompt Template 3:*** *"I have performed scRNA-Seq clustering on a human tumor biopsy and identified 8 clusters. The top 10 marker genes for each cluster are listed below. Please annotate each cluster with the most likely cell type, confidence level (High/Medium/Low), key supporting markers, and reasoning. Also flag any clusters that might represent doublets. Cluster 0: EPCAM, KRT19, KRT8, MUC1, CDH1, CLDN4, TACSTD2, KRT18, GRHL2, ELF3. Cluster 1: CD3E, CD8A, GZMB, PRF1, NKG7, GNLY, IFNG, CCL4, CXCR3, LAG3. Cluster 2: CD68, CD14, CSF1R, MRC1, CD163, APOE, C1QA, C1QB, TREM2, SPP1. Cluster 3: COL1A1, COL3A1, VIM, FAP, ACTA2, FN1, PDGFRA, THY1, DCN, LUM. Cluster 4: PECAM1, VWF, CDH5, CLDN5, ESAM, ENG, EMCN, ROBO4, FLT1, KDR. Cluster 5: CD19, MS4A1, CD79A,*

> *CD79B, IGHM, IGHD, BANK1, CD22, BLK, PAX5. Cluster 6: NCAM1, KLRD1, KLRB1, FCGR3A, GNLY, NKG7, GZMB, PRF1, TYROBP, FCER1G. Cluster 7: KIT, TPSAB1, TPSB2, CPA3, HDC, MS4A2, FCER1A, GATA2, CD63, SLC18A2."*

```
library(Seurat); library(ggplot2); library(dplyr)
```

```
# Load scType functions
source("https://raw.githubusercontent.com/IanevskiAleksandr/sc-type/master/R/gene_sets_prepare.R")
source("https://raw.githubusercontent.com/IanevskiAleksandr/sc-type/master/R/sctype_score_.R")

# Load marker gene database
gs_list <- gene_sets_prepare(
 "https://raw.githubusercontent.com/IanevskiAleksandr/sc-type/master/ScTypeDB_full.xlsx",
 "Immune system")  # Change to match your tissue type

seurat_filt <- readRDS("tumor_seurat.rds")
scRNAseqData <- GetAssayData(seurat_filt, assay="SCT", slot="scale.data")

# Calculate scType scores
es_max <- sctype_score(scRNAseqData=scRNAseqData, scaled=TRUE,
 gs=gs_list$gs_positive, gs2=gs_list$gs_negative)

# Assign cell type per cluster
cL_results <- do.call("rbind", lapply(unique(seurat_filt$seurat_clusters), function(cl) {
 es_max_cl <- sort(rowSums(es_max[,rownames(seurat_filt@meta.data[
  seurat_filt@meta.data$seurat_clusters==cl,])]), decreasing=TRUE)
 head(data.frame(cluster=cl, type=names(es_max_cl), scores=es_max_cl,
            ncells=sum(seurat_filt@meta.data$seurat_clusters==cl)), 10)
}))
sctype_scores <- cL_results %>% group_by(cluster) %>% top_n(n=1, wt=scores)
sctype_scores$type[as.numeric(sctype_scores$scores) < sctype_scores$ncells/4] <- "Unknown"
cat("scType annotations:\n"); print(sctype_scores %>% select(cluster,type,scores,ncells))

seurat_filt[["sctype_annotation"]] <- sctype_scores$type[
 match(seurat_filt$seurat_clusters, sctype_scores$cluster)]

# Prepare marker gene table for LLM annotation
markers <- FindAllMarkers(seurat_filt, only.pos=TRUE, min.pct=0.25, logfc.threshold=0.25)
top10_for_llm <- markers %>% group_by(cluster) %>%
 slice_max(order_by=avg_log2FC, n=10) %>%
 summarise(top_markers=paste(gene, collapse=", "), .groups="drop")
cat("\n=== COPY TO LLM (GPT-4/Claude/DeepSeek) ===\n")
```

```
for (i in seq_len(nrow(top10_for_llm)))
  cat(sprintf("Cluster %s: %s\n", top10_for_llm$cluster[i], top10_for_llm$top_markers[i]))

# After LLM annotation, enter results here and compare
llm_annotations <- c("0"="Epithelial/Cancer cells","1"="CD8+ Cytotoxic T cells",
  "2"="Tumor-associated Macrophages","3"="Cancer-associated Fibroblasts",
  "4"="Endothelial cells","5"="B cells","6"="NK cells","7"="Mast cells")
seurat_filt[["llm_annotation"]] <- llm_annotations[as.character(seurat_filt$seurat_clusters)]

p1 <- DimPlot(seurat_filt, group.by="sctype_annotation", label=TRUE, label.size=3) +
  labs(title="scType Automated Annotation") + NoLegend()
p2 <- DimPlot(seurat_filt, group.by="llm_annotation", label=TRUE, label.size=3) +
  labs(title="LLM-Assisted Annotation (GPT-4)") + NoLegend()
p1 + p2 + plot_annotation(title="Cell Type Annotation: scType vs. LLM-Assisted",
  theme=theme(plot.title=element_text(face="bold")))
```

2.5 Cutting-Edge: Multimodal Single-Cell Omics and the Human Cell Atlas

CITE-seq simultaneously measures mRNA expression and surface protein abundance in the same single cell using **antibody-derived tags (ADTs)** — antibodies conjugated to unique oligonucleotide barcodes, captured and sequenced alongside mRNA. The **Seurat v4 WNN (Weighted Nearest Neighbor)** algorithm integrates RNA and protein modalities by learning the optimal weight for each modality per cell [10].

10x Multiome (ATAC + RNA) simultaneously measures chromatin accessibility (ATAC-seq) and gene expression (RNA-seq) in the same single nucleus. This enables **cis-regulatory analysis**: linking specific enhancers (ATAC peaks) to their target genes (correlated RNA expression), revealing transcription factor networks driving cell type identity. A key application is linking GWAS variants (in non-coding regulatory regions) to their target genes in specific cell types.

The Human Cell Atlas (HCA) is an international consortium launched in 2016 aiming to create a comprehensive reference map of all ~37 trillion human cells. As of 2024, the HCA has profiled over **50 million cells** across more than 60 tissues, with published draft atlases for the lung, gut, kidney, liver, brain, immune system, and developing embryo. HCA data is publicly available through the **CellxGene portal** (cellxgene.cziscience.com), providing interactive visualization and download of >700 datasets.

Technology	Modalities Measured	Resolution	Key Software	Best Application
CITE-seq	RNA + Surface proteins (ADTs)	Single cell	Seurat v4 (WNN), Scanpy	Immune cell phenotyping; precise cell type ID

Technology	Modalities Measured	Resolution	Key Software	Best Application
10x Multiome (ATAC+RNA)	Chromatin accessibility + RNA	Single nucleus	Seurat, ArchR, Signac	Regulatory genomics; cis-regulatory analysis
Paired-seq	Chromatin accessibility + RNA	Single cell	Custom	Regulatory genomics
SHARE-seq	Chromatin accessibility + RNA	Single cell	Seurat, ArchR	Hair follicle differentiation; skin biology
Spatial CITE-seq	RNA + Proteins + Spatial coordinates	Spot (~1–10 cells)	Seurat	Spatial immune profiling; tumor microenvironment
CODEX/PhenoCycler	>50 proteins + Spatial coordinates	Single cell	CODEX Processor, Squidpy	Spatial proteomics; tissue architecture

Table 6. Multimodal single-cell omics technologies: modalities measured, resolution, key software, and best application 10.

R Code Block 4 — Pseudobulk Differential Expression Analysis (scRNA-Seq → DESeq2)

> ***LLM Prompt Template 4:*** *"I have a Seurat object 'tumor_seurat.rds' with scRNA-Seq data from 5 COVID-19 patients and 5 healthy controls (metadata: 'sample_id', 'condition' [COVID/Healthy], 'cell_type'). Please write R code for pseudobulk differential expression analysis comparing COVID-19 vs. Healthy for each cell type using DESeq2. Include: (1) pseudobulk aggregation per sample per cell type; (2) filter cell types with <3 cells per sample; (3) DESeq2 with design ~condition; (4) BH-FDR correction; (5) volcano plot per cell type; (6) heatmap of top 30 DE genes; (7) save results as CSV. Explain in comments why pseudobulk is statistically correct and why single-cell DE is not appropriate for multi-sample comparisons."*

```
# WHY PSEUDOBULK?
```

```
# Running DE directly on single cells (FindMarkers with group.by="condition")
# treats each cell as an independent observation. But cells from the same patient
# are NOT independent — they share genetic background, environment, and processing.
# This inflates sample size from n=5 patients to n=5,000 cells → thousands of
# false positives. Pseudobulk aggregates cells per sample per cell type, then
# runs DESeq2 on n=5 samples per group — statistically valid.

library(Seurat); library(DESeq2); library(ggplot2); library(dplyr)
library(pheatmap); library(tidyverse)

seurat_obj <- readRDS("tumor_seurat.rds")
cell_types <- unique(seurat_obj$cell_type)
all_results <- list()

for (ct in cell_types) {
```

```
  ct_cells <- subset(seurat_obj, subset=cell_type==ct)
  cells_per_sample <- table(ct_cells$sample_id)
  valid_samples <- names(cells_per_sample[cells_per_sample >= 3])
  if (length(valid_samples) < 6) {
   cat(sprintf("Skipping %s: insufficient cells\n", ct)); next
  }
  ct_cells <- subset(ct_cells, subset=sample_id %in% valid_samples)

  # Aggregate raw counts per sample (pseudobulk)
  pb_counts <- AggregateExpression(ct_cells, assays="RNA",
   return.seurat=FALSE, group.by="sample_id")$RNA

  sample_meta <- data.frame(
   sample_id=colnames(pb_counts),
   condition=factor(ifelse(grepl("COVID",colnames(pb_counts)),"COVID","Healthy"),
                    levels=c("Healthy","COVID")),
   row.names=colnames(pb_counts))

  dds <- DESeqDataSetFromMatrix(countData=round(pb_counts),
   colData=sample_meta, design=~condition)
  keep <- rowSums(counts(dds) >= 10) >= 3
  dds <- dds[keep,]
  dds <- DESeq(dds, quiet=TRUE)

  res_df <- as.data.frame(results(dds, contrast=c("condition","COVID","Healthy"),
   alpha=0.05)) %>%
   rownames_to_column("gene") %>%
   mutate(cell_type=ct,
          sig=!is.na(padj) & padj < 0.05 & abs(log2FoldChange) > 1,
          direction=case_when(sig & log2FoldChange > 0 ~ "Up in COVID",
                              sig & log2FoldChange < 0 ~ "Down in COVID",
                              TRUE ~ "Not significant")) %>%
   arrange(padj)
  all_results[[ct]] <- res_df
  cat(sprintf("%s: %d up, %d down\n", ct,
   sum(res_df$direction=="Up in COVID",na.rm=TRUE),
   sum(res_df$direction=="Down in COVID",na.rm=TRUE)))
}

# Volcano plots per cell type
volcano_plots <- lapply(names(all_results), function(ct) {
  df <- all_results[[ct]]
```

```
  top_labels <- df %>% filter(sig) %>% arrange(padj) %>% head(8)
  ggplot(df, aes(x=log2FoldChange, y=-log10(padj+1e-300), color=direction)) +
    geom_point(size=1.5, alpha=0.6) +
    scale_color_manual(values=c("Up in COVID"="#E53935","Down in COVID"="#1565C0",
                                "Not significant"="#90A4AE")) +
    geom_hline(yintercept=-log10(0.05), linetype="dashed", color="#FF8F00") +
    geom_vline(xintercept=c(-1,1), linetype="dotted", color="#546E7A") +
    geom_text(data=top_labels, aes(label=gene), size=2.5, hjust=-0.1, color="#1a237e") +
    theme_bw() +
    labs(title=paste0(ct,"\n(Pseudobulk DESeq2: COVID vs. Healthy)"),
         x="log2 Fold Change", y="-log10(Adjusted p-value)") +
    theme(plot.title=element_text(face="bold",size=9), legend.position="bottom")
})
library(patchwork); wrap_plots(volcano_plots, ncol=3)

# Heatmap of top DE genes across cell types
top_genes <- bind_rows(all_results) %>% filter(sig) %>%
  group_by(cell_type) %>% slice_min(order_by=padj, n=5) %>% pull(gene) %>% unique()
lfc_matrix <- sapply(names(all_results), function(ct) {
  df <- all_results[[ct]]
  lfc <- df$log2FoldChange[match(top_genes, df$gene)]
  lfc[is.na(lfc)] <- 0; return(lfc)
})
rownames(lfc_matrix) <- top_genes
pheatmap(lfc_matrix,
  color=colorRampPalette(c("#1565C0","white","#E53935"))(100),
  breaks=seq(-4,4,length.out=101), cluster_rows=TRUE, cluster_cols=TRUE,
  fontsize_row=8, fontsize_col=9,
  main="Top DE Genes per Cell Type\n(log2FC: COVID vs. Healthy, Pseudobulk DESeq2)")

all_results_combined <- bind_rows(all_results)
write.csv(all_results_combined, "pseudobulk_DE_results_all_celltypes.csv", row.names=FALSE)
cat("Saved: pseudobulk_DE_results_all_celltypes.csv\n")
```

```
cat("Total significant DE genes:", sum(all_results_combined$sig, na.rm=TRUE), "\n")
```

> ⚠️ ***LLM Hallucination Warning — Single-Cell & Spatial Transcriptomics-Specific Pitfalls***

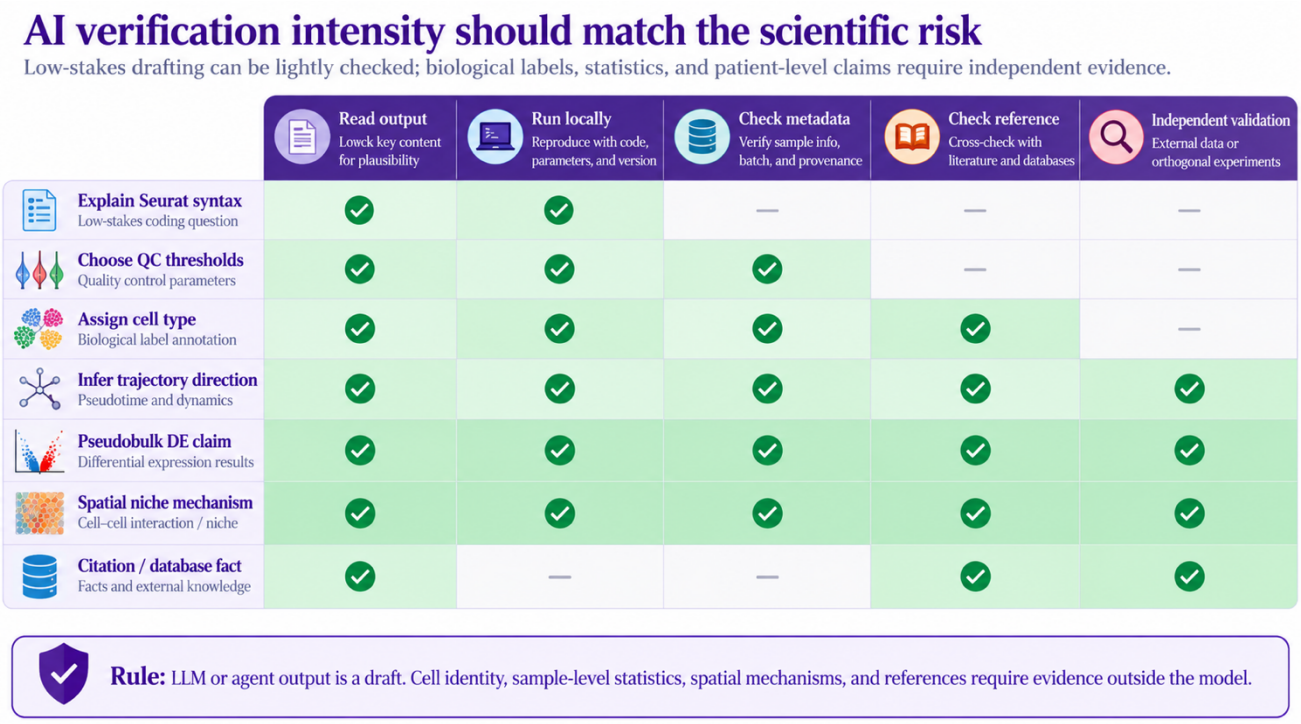

Figure 14. AI verification matrix. The required verification intensity rises with the scientific consequences of the output.

Error Type	Hallucinated Code / Statement	Correct Approach	Consequence
Double normalization	NormalizeData() after SCTransform()	Use either NormalizeData + ScaleData OR SCTransform — never both	Double-normalization distorts expression values
Single-cell DE for multi-sample	FindMarkers(group.by="condition") directly on cells	Pseudobulk with DESeq2 per cell type	Inflated sample size → thousands of false positives
Ignoring mitochondrial QC	No percent.mt filter	Filter cells with percent.mt > 20–25%	Dead/stressed cells contaminate all downstream analyses
Over-interpreting UMAP distances	"Cluster A is more similar to B because they are closer in UMAP"	UMAP distances are not quantitatively meaningful; only cluster membership is reliable	Incorrect biological conclusions from UMAP layout
t-SNE for trajectory analysis	Running Monocle3 on t-SNE coordinates	Always use UMAP coordinates for trajectory analysis	t-SNE does not preserve global structure; trajectories are meaningless

Error Type	Hallucinated Code / Statement	Correct Approach	Consequence
Skipping doublet detection	No DoubletFinder/Scrublet step	Run DoubletFinder or Scrublet before clustering	Doublets appear as "intermediate" cell types; mislead trajectory analysis
Wrong resolution parameter	FindClusters(resolution=0.8) for all datasets	Tune resolution; validate with known markers; use clustree	Over- or under-clustering; incorrect cell type assignments
Confusing Visium spot with single cell	"Each Visium spot = one cell"	Each Visium spot (55 µm) contains 1–10 cells; requires deconvolution	Incorrect cell type proportion estimates
Normalized counts in DESeq2	DESeqDataSetFromMatrix(countData=NormalizeData(counts))	DESeq2 requires raw integer UMI counts; it performs its own normalization	DESeq2 size factor estimation fails; incorrect DE results
Misinterpreting pLDDT as UMAP confidence	"UMAP coordinates represent cell type assignment confidence"	UMAP is a visualization; confidence comes from cluster marker gene enrichment scores	Misinterpretation of cell type assignment reliability

Key Concepts Summary

Practical Decision Table: What Should AI Do - and What Must the Student Verify?

Task	Useful AI / agent role	Mandatory human verification
QC planning	Suggest metrics and plots; identify likely doublet/ambient-RNA checks.	Thresholds should be tissue- and chemistry-aware; inspect distributions per sample.
Code generation	Draft Seurat/Scanpy/DESeq2 code and explain functions.	Run locally; inspect object dimensions, warnings, package versions, and assays/layers.
Cell annotation	Generate candidate labels and supporting/contradictory markers.	Validate with tissue-matched reference, negative markers, independent classifier, and expert review.
Trajectory	Suggest Monocle3/scVelo workflow and candidate roots.	Root choice and kinetic assumptions are biological; confirm direction with known time/lineage evidence.
Pseudobulk DE	Aggregate counts and construct design/contrast.	Sample is replicate; use raw integer counts; verify donor-condition mapping and coefficient direction.
Spatial domains	Propose Nicheformer/Novae/BANKSY/GraphST analyses.	Confirm domains with morphology, known anatomy, replicate sections, and marker localization.
Cell-cell communication	Rank ligand-receptor hypotheses.	Expression + proximity are not causality; validate receptor context and perturbation evidence.
Writing	Draft methods/results from verified objects.	Never invent n, p, software version, accession, or citation; regenerate text after final analysis.

Concept	Definition	Practical Application
scRNA-Seq	Single-cell RNA sequencing — measures gene expression in individual cells	Resolves cellular heterogeneity invisible in bulk RNA-Seq
GEM	Gel Bead in Emulsion — oil droplet containing one cell + one barcoded gel bead (10x Chromium)	The fundamental unit of 10x scRNA-Seq; enables cell barcode + UMI tagging
Cell barcode	Unique 16 bp DNA sequence identifying all cDNA from a single cell	Enables computational assignment of reads to cells after pooled sequencing
UMI	Unique Molecular Identifier — 12 bp random sequence per mRNA molecule	Enables PCR duplicate removal; provides absolute molecule counting

Concept	Definition	Practical Application
nFeature_RNA	Number of unique genes detected per cell	QC metric: <200 = empty droplet; >6,000 = likely doublet
percent.mt	Percentage of reads mapping to mitochondrial genes	QC metric: >20–25% = dying/stressed cell; filter before analysis
Doublet	Two cells captured in one GEM; appears as one cell with inflated transcriptome	Detected by DoubletFinder/Scrublet; must be removed before clustering
SCTransform	Regularized negative binomial regression normalization	Recommended for datasets with high sequencing depth variation; replaces NormalizeData
HVG	Highly Variable Gene — gene with high cell-to-cell expression variation	Top 2,000–5,000 HVGs used for PCA; capture biologically meaningful variation
UMAP	Uniform Manifold Approximation and Projection — 2D visualization of PCA space	Visualization only; distances not quantitatively meaningful; only cluster membership is reliable
Leiden/Louvain	Graph-based community detection algorithms for cell clustering	FindClusters() in Seurat; resolution parameter controls granularity
Cell type annotation	Assignment of biological identity to each cluster based on marker genes	Manual (expert), automated (CellTypist, scType), or LLM-assisted (GPT-4/Claude)
Pseudotime	Computational ordering of cells along a developmental trajectory	Monocle3; assigns each cell a value from 0 (stem) to 1 (differentiated)
RNA velocity	Prediction of future cell state from unspliced/spliced transcript ratios	scVelo; visualized as arrows on UMAP; predicts differentiation direction
Pseudobulk DE	Aggregating counts per cell type per sample before running DESeq2	Statistically correct approach for multi-sample scRNA-Seq DE analysis
Spatial transcriptomics	Measurement of gene expression with spatial coordinates preserved	10x Visium (55 µm spots), Stereo-seq (500 nm), MERFISH (single cell)
Spot deconvolution	Estimating cell type proportions within each Visium spot	RCTD, cell2location, Tangram; requires paired scRNA-Seq reference
SVG	Spatially Variable Gene — gene with spatially structured expression pattern	SpatialDE, SPARK-X, Moran's I; spatial equivalent of differentially expressed gene
CITE-seq	Simultaneous measurement of RNA + surface proteins in single cells	Antibody-derived tags (ADTs); Seurat v4 WNN integration
scGPT	Transformer foundation model pre-trained on 33M single-cell transcriptomes	Fine-tuned for annotation, batch correction, perturbation prediction
Geneformer	Transformer model pre-trained on 30M cells using gene rank-value encoding	In silico perturbation prediction; identified novel cardiomyopathy targets
Human Cell Atlas	International consortium mapping all ~37 trillion human cells	>50M cells profiled; CellxGene portal; reference for automated annotation

Practice Questions

1. Explain why bulk RNA-Seq cannot detect rare cell populations. A tumor biopsy contains 98% cancer cells and 2% tumor-infiltrating T cells. If you perform bulk RNA-Seq, what will the expression profile look like? What would scRNA-Seq reveal that bulk RNA-Seq cannot? Give a specific clinical example where this distinction matters.

2. What is a cell barcode? What is a UMI? Why are UMIs essential for accurate gene expression quantification in scRNA-Seq? If a gene is detected 10,000 times in a cell's reads but has only 25 unique UMIs, what does this tell you about the true expression level of that gene, and what would happen to your analysis if you did not use UMI deduplication?

3. A scRNA-Seq dataset has the following cell-level QC metrics: Cell A: nFeature=350, nCount=500, percent.mt=2%; Cell B: nFeature=8,500, nCount=18,000, percent.mt=1.5%; Cell C: nFeature=150, nCount=200, percent.mt=48%; Cell D: nFeature=2,500, nCount=4,000, percent.mt=3%; Cell E: nFeature=3,200, nCount=5,500, percent.mt=22%. Which cells would you keep and which would you filter out? Justify each decision with the biological interpretation of each metric.

4. Explain the difference between NormalizeData() and SCTransform() in Seurat. What specific technical artifact does SCTransform address that library size normalization does not? Why is it important never to apply both normalizations to the same dataset? Under what circumstances would you prefer NormalizeData() over SCTransform()?

5. Describe the graph-based clustering approach used in Seurat. What is the resolution parameter and how does it affect the number of clusters? After clustering, you identify a cluster with top marker genes: CD3E, CD8A, GZMB, PRF1, LAG3, PDCD1. What cell type is this, and what is its functional state? How would you use an LLM to assist with cell type annotation, and what are the risks of relying solely on LLM annotation without validation?

6. You have a scRNA-Seq dataset with 5 COVID-19 patients and 5 healthy controls, with ~5,000 cells per sample. A colleague runs FindMarkers(group.by="condition") directly on all T cells and reports 3,500 differentially expressed genes. You suspect this is statistically invalid. Explain why this approach is incorrect and describe the correct pseudobulk approach using DESeq2. What is the effective sample size in each approach, and why does this matter?

7. Compare 10x Visium and MERFISH. What is the fundamental difference in their measurement principle (sequencing-based vs. imaging-based)? What are the tradeoffs in terms of gene coverage, spatial resolution, and cost? For each of the following research questions, specify which platform you would choose and why: (a) mapping the spatial distribution of all ~20,000 genes in a human liver biopsy; (b) measuring the precise subcellular localization of 500 specific genes in intact brain tissue at single-cell resolution.

8. Explain the concept of RNA velocity. What is the biological basis for using unspliced/spliced transcript ratios to predict future cell states? How is RNA velocity visualized on a UMAP plot? Give one specific biological application where RNA velocity provides information that pseudotime analysis alone cannot. What additional data processing step is required before RNA velocity analysis that is not needed for standard scRNA-Seq analysis?

9. What is scGPT and how does it differ from using a general-purpose LLM (like GPT-4) for single-cell analysis? What does "pre-training on 33 million single-cell transcriptomes" mean in practice — what kind of biological knowledge does the model learn? Name three specific tasks that scGPT can

perform that a general LLM cannot, and explain why the domain-specific pre-training is necessary for each task.

10. (Integrative Design Question) You are studying the gut microbiome-liver axis in NAFLD, connecting to Dr. Xie's research program. You have: (1) 16S microbiome data from 20 NAFLD patients and 20 healthy controls (Week 6); (2) bulk liver transcriptomics from the same patients (Week 5); (3) plasma metabolomics including TMAO, BCAAs, and bile acids (Week 7); and now (4) single-cell RNA-Seq of liver biopsies from 5 NAFLD patients and 5 healthy controls. Design a multi-omics study integrating all four data types. For each data type, specify: what specific biological question it answers; what cell type in the liver you would expect to show the strongest transcriptional response to gut dysbiosis; and how you would use the pseudobulk DE results from scRNA-Seq (Code Block 4) in the Week 9 multi-omics integration pipeline.

Suggested Reading List

Additional AI / Agent Practice Questions

1. Why is a cluster-level marker list insufficient to prove a cell identity? Name four independent evidence sources an AI agent should check.

2. Explain why cells are not biological replicates in a multi-patient scRNA-seq study. What does pseudobulk preserve that cell-level testing ignores?

3. A foundation model predicts that a cell belongs to a fibrotic macrophage niche. What measurements would you require before treating this as a biological conclusion?

4. Compare scGPT/scFoundation with Nicheformer/Novae. Which models explicitly encode spatial context, and why does that matter?

5. Design a verifier-agent checklist for an analysis that transfers scRNA-seq cell labels onto a Visium HD or Xenium dataset.

6. Distinguish the claims: spatial colocalization, ligand-receptor compatibility, inferred communication, and causal signaling. What additional evidence is required at each step?

One-Hour Homework: AI-Assisted Single-Cell + Spatial Evidence

Goal: use AI as an assistant while demonstrating that the student can independently verify sample identity, cell annotation, replicate-aware statistics, and spatial interpretation. The homework is intentionally smaller than a full pipeline and is designed to prepare for a short Week 9 quiz.

0-10 min: Inspect a provided Seurat metadata table: sample_id, condition, cluster, nFeature_RNA, nCount_RNA, percent.mt. Identify one QC concern and confirm that cells are nested within samples.

10-20 min: For two selected clusters, give an LLM the top markers using CARE+V. Ask for 3 candidate labels, supporting markers, contradictory markers, and references/databases to check.

20-35 min: Aggregate one cell type by sample and run a single pseudobulk comparison in DESeq2. Record the exact contrast and number of biological samples per group.

35-50 min: Use a small spatial table (x, y, cell_type, marker expression) to make one spatial plot and test a simple neighborhood/colocalization question.

50-60 min: Write a 150-word interpretation with one supported claim, one claim not yet supported, one limitation, and an AI verification log.

> *Required submission: R script; cell-annotation evidence table; pseudobulk result CSV; one spatial figure; 150-word verification note; exact AI/agent prompt and verification log; sessionInfo().*

Foundational (read before class)

#	Reference	Why Read It
1	Macosko et al. (2015) *Cell* 161:1202–1214	Drop-seq: the first droplet-based scRNA-Seq method; establishes the conceptual framework for cell barcoding and UMIs
2	Stuart et al. (2019) *Cell* 177:1888–1902	Seurat v3: the standard scRNA-Seq analysis framework; must-read for understanding the complete Seurat workflow
3	Ståhl et al. (2016) *Science* 353:78–82	Original spatial transcriptomics paper (basis for 10x Visium); Nobel Prize-adjacent work establishing the field

Core Methods Papers (read alongside handout)

#	Reference	Key Finding
4	Hafemeister & Satija (2019) *Genome Biology* 20:296	SCTransform: the recommended normalization method for scRNA-Seq; addresses sequencing depth confounding
5	Bergen et al. (2020) *Nature Biotechnology* 38:1408–1414	scVelo: RNA velocity with dynamical modeling; enables prediction of future cell states
6	Cao et al. (2019) *Nature* 566:496–502	Monocle 3: trajectory analysis and pseudotime; the standard tool for developmental trajectory reconstruction
7	Cable et al. (2022) *Nature Biotechnology* 40:317–326	RCTD: robust cell type deconvolution for spatial transcriptomics; the standard deconvolution tool
8	Hao et al. (2021) *Cell* 184:3573–3587	Seurat v4 / WNN: multimodal single-cell analysis integrating RNA and protein (CITE-seq)

Clinical/Translational (read for clinical context)

#	Reference	Key Finding
9	Regev et al. (2017) *eLife* 6:e27041	The Human Cell Atlas: vision, goals, and progress; the most ambitious single-cell project in history
10	Sikkema et al. (2023) *Nature Medicine* 29:1563–1577	Human Lung Cell Atlas: 2.4 million cells across 486 individuals; landmark clinical application of scRNA-Seq
11	Guilliams et al. (2022) *Cell* 185:379–396	Spatial proteogenomics reveals distinct and evolutionarily conserved hepatic macrophage niches; liver zonation

Cutting-Edge / AI (optional, for deeper exploration)

#	Reference	Key Finding
12	Cui et al. (2024) *Nature Methods* 21:1117–1128	scGPT: foundation model for single-cell biology pre-trained on 33M cells
13	Theodoris et al. (2023) *Nature* 618:616–624	Geneformer: in silico perturbation prediction from single-cell foundation model; novel cardiomyopathy targets
14	Chen et al. (2022) *Nature Biotechnology* 40:163–173	Spatially resolved multiomics of single cells in tissue; CITE-seq + spatial integration
15	Kleshchevnikov et al. (2022) *Nature Biotechnology* 40:661–671	cell2location: Bayesian model for spatial cell type deconvolution; more accurate than RCTD for rare cell types

---

Pre-Reading for Week 9

Before the next class on **Multi-Omics Integration & Pathway Enrichment (KEGG/GO)** (28 October 2026), please:

1. Review the DESeq2 results format from Week 5 (bulk RNA-Seq) and the pseudobulk DE results from Code Block 4 (this week) — both produce the same output format (gene, log2FoldChange, padj) and will be directly integrated in Week 9.
2. Review the clusterProfiler pathway enrichment workflow from Week 7 (Code Block 1, Step 9) — Week 9 will extend this to multi-omics integration using MOFA+ (Multi-Omics Factor Analysis).
3. Install the following R packages: install.packages(c("MOFA2","mixOmics","ggplot2","ComplexHeatmap")).
4. **Conceptual connection to Week 8:** The scRNA-Seq cell type proportions estimated by RCTD deconvolution (Code Block 2) can be used to deconvolve bulk RNA-Seq data — estimating the cell type composition of bulk samples using scRNA-Seq as a reference. This is a key multi-omics integration strategy that will be covered in Week 9.
5. Prof. Guangchuang Yu's **clusterProfiler** package will be the primary tool for pathway enrichment in Week 9. Review the clusterProfiler documentation at https://bioconductor.org/packages/clusterProfiler.

References

Updated 2025-2026 AI / Agent References

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18. Tejada-Lapuerta A, et al. Nicheformer: a foundation model for single-cell and spatial omics. Nature Methods. 2025;22:2525-2538. doi:10.1038/s41592-025-02814-z.

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23. STAT: A multi-agent framework for integrated and interactive spatial transcriptomics analysis. bioRxiv preprint. 2026. doi:10.64898/2026.05.01.722244.

24. 10x Genomics. Cell Ranger Gene Expression Algorithm, current documentation (Cell Ranger v10.1, accessed 2026): cell calling uses OrdMag and EmptyDrops-based steps.

25. 10x Genomics. Xenium Prime 5K panels and Visium HD / Space Ranger current documentation, accessed 2026.